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Is ICE freezing US agriculture? Farm-level adjustment to increased local immigration enforcement☆

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ABSTRACT

We study the impact of US immigration enforcement through a locally-implemented program known as 287(g) on agriculture using confidential farm-level data and farm census data. We address the endogeneity of program participation by using pre-program county jail occupancy as an instrument. We provide evidence that the agricultural sector experienced limited endogenous technical change in response to the negative labor supply shock caused by increased immigration enforcement. Fuel expenses increased, consistent with predictions of an increase in automated technology by models of endogenous technical change. However, a corresponding decline in agricultural acres and employment and an increase in labor expenditure suggests capital-labor substitution was insufficient to fully offset the impact of the shock.

1. Introduction

Many industries rely on immigrants, both documented and undocumented, for their labor force; the impact of current immigration policies and labor force uncertainty on workers and businesses is the subject of heated debate in many countries, i.e. Akbari and MacDonald (2014); Dancygier and Margalit (2020); Hovden and Mjelde (2019); Kerr et al. (2011). Theoretically, the effect of restricting immigration will depend on whether domestic labor and/or mechanization can substitute for immigrant labor. Empirically, there has been little documentation of the technical changes that may be induced by labor scarcity caused by immigration restrictions, with few exceptions, notably Clemens et al. (2018). The effects of immigration policy on labor and capital substitution are difficult to measure because of data constraints, as well as the challenge of disentangling national immigration policy from broader economic trends and local policies from local institutional factors. While there is a large literature that studies the impacts of immigration policies on workers and wages (e.g., Borjas, 1987; Chiswick, 1978; Ottaviano and Peri, 2012)), less is known about the impacts of immigration policies on firms, where the technical changes would be observed, and especially in sectors, like agriculture, that are heavily reliant on migrant labor.

This paper documents how US farm businesses were affected by a localized change in immigration enforcement intensity, using farm-level unbalanced panel data and county-level Census of Agriculture aggregates. We use the immigration policy-induced labor supply shock to test for evidence of endogenous technological change in the agricultural sector. The immigration program in question, commonly called the 287(g) program, authorized law enforcement entities to function as US Immigration and Custom Enforcement (ICE) agents in order to identify and detain unauthorized immigrants within their jurisdiction as part of their routine duties. These entities include Sheriff's Offices and Police Departments in 13 US cities and 49 counties across 20 states. We focus on 287(g) implementation at the local, county level from 2007 to 2012. This program not only directly affected the supply of immigrant workers but also served as a deterrent that caused migrants to avoid participating areas (Dee and Murphy, 2020).

This study takes advantage of the spatial and temporal variation of 287(g) program adoption to estimate how US farm businesses adapted to increased local immigration enforcement. We use two *different* sets of agricultural survey panel data, one firm- and one county-level, which allows us to examine farms that were randomly sampled both before and after the policy change and the local agricultural sector, respectively. We evaluate evidence for endogenous technical change using outcomes that measure farm- and county-level fuel expenditure, labor expenditure, and acres operated, as well as county-level farm employment. These dependent variables provide insight into how employers adjusted labor and capital use following 287(g) authorization, and they were selected to test some of the implications of the model of endogenous technical change in farm sector developed by Clemens et al. (2018). Due to data limitations, we are not able to consider wage impacts, directly estimate labor sup-

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ply or other elasticities, or identify changes in the production of specific specialty crops that are more easily mechanized.

Causal identification of labor supply shocks is a major challenge to understanding the impact of immigration. Generally, this endogeneity has been overcome in two ways: either by exploiting an unexpected exogenous shock, like the Mariel boatlift (Card, 1990), or by instrumenting for the current immigrant population using the past populations of immigrants from the same place of origin, relying on the enclaving tendency of immigrants (Card, 2009). The former occurs infrequently and typically only in one place, affecting external validity. The latter is usually used for urban areas in the US, as such enclaves are more prevalent there. This does little to help inform the debate in agriculture, which is one of the largest employers of migrant laborers, who typically move from one job-site to another.

We implement a novel and robust instrumental variables strategy for local immigration enforcement decisions: we use measures of local jail occupancy before 287(g) authorization, which we argue are unrelated to agricultural production decisions, as instrumental variables for the decision to participate in 287(g). Aggregated up to the county level, local jail occupancy prior to the start of the program is a strong predictor of 287(g) participation, with little evidence of any relationship to farm sector activity. Instead, jail occupancy in program counties appears to be related to increased jail capacity expansion through renovations of existing facilities or construction of new ones in the decade prior to 287(g) implementation. There is no evidence that population growth (or decline) drove this expansion, which we argue is indicative of active and well-supported law enforcement where investment in jail facilities outpaced jail population. There was a widespread ex-ante belief that 287(g) participation could increase local law enforcement jurisdictions' revenue (Shahani and Greene, 2009). This belief, along with the political popularity of anti-immigrant action, drove program participation and so underpins our first stage; its validity is not influenced by ex-post anecdotes that this revenue was not realized[1]. Further, farm and (relevant) non-farm sector trends appear to be largely unrelated to future 287(g) participation. This approach allows us to provide one the first plausibly causal estimates of the economic impact of local immigration policy on US firms.

We find evidence that increased enforcement of 287(g) led to large declines in employment and an increase in mechanization. We observe a decline in number of workers employed and a local increase in labor expenditure, which suggests some combination of wage increases and a reallocation to more expensive labor, which typically would include machinery operators. Farms increased fuel expenditures (a proxy for higher use of automated technologies), but agricultural land use in the county decreased. With respect to endogenous technical change, we conclude that both full labor and capital substitution was not feasible, given the concurrent large declines in employment and land use. The increase in county-level labor expenditure further suggests non-marginal changes in farm structure. Our results are in line with concerns that more stringent immigration enforcement disadvantage sectors that rely on manual or immigrant labor.

Our paper builds on several related literatures. Most closely related is Clemens et al. (2018), which provides a theoretical framework and empirical tests for endogenous technical change in farms due to the abrupt end of a large US national agricultural guest-worker program (the bracero exclusion) in 1964 and state-level variation in bracero employment/program participation. The outcomes we consider are not directly comparable with Clemens et al. (2018), and we observe changes to local immigration policy in the modern farm economy instead of national policy in the 1960s. Given these caveats, our results are generally consistent while suggesting lower substitutability of farm labor and capital. Clemens et al. (2018) finds that the bracero exclusion policies did not increase employment of domestic farm workers and documents a short term decline in acres harvested of "easily-mechanized bracero-intensive crops" and a longer term decline of those that were less easily mechanized. We find an increase in fuel expenditure consistent with automation, but also evidence of contracting production. We build on papers that have studied the 287(g) program with a focus on employment and urban areas: Kostandini et al. (2014) and Dee and Murphy (2020) document that the program led to a decline in the population of non-citizens in program jurisdictions, and Bohn and Santillano (2017) show that employment impacts varied heavily across industries.

With this paper, we contribute to ongoing policy debates that are supported by thin empirical evidence on how firms' non-wage decisions are affected by a fluctuating labor supply (Olney, 2013). Our approach allows us to provide evidence on the substitutabilty of native workers and/or capital for immigrant workers in a sector heavily dependent on an immigrant labor force. Foreign-born workers tend to specialize in manual labor, which can limit their substitutability (Peri and Sparber, 2009). The extent to which this substitutability exists, if it does at all, is a crucial point of difference in many arguments about the direction US immigration policy should take, and plays a key role in theoretically determining firm-level impacts of immigration policy (Lewis, 2003).

In the next section we provide conceptual and institutional background that underpins our study. Then we describe our data and empirical model, including our instrumental variables strategy. This is following by a discussion of our results, including several robustness checks. In the conclusion, we summarize our findings and draw policy implications.

2. Background

In this section, we discuss (1) predictions for the impact of labor supply shocks on endogenous technical change, based on findings of related literature and (2) relevant background on the role of immigrant labor in the US farm economy and the 287(g) program.

2.1. Labor supply shocks and endogenous technical change

Under theoretical models of endogenous technical change, a decrease in the supply of one type of labor causes substitution away from that type. If production technology can always adapt to the available labor supply, different industries will be largely unaffected by policy changes that change the relative availability of a "type" of worker. The underlying assumption of adaptive technological change may be violated in industries, like agriculture, that rely on manual labor. In examining the sector-level response to labor supply shocks, Lewis (2003) finds that such shocks do not affect the local sector mix. Instead, increases in the supply of a type of labor, such as manual immigrant labor, tend to increase the relative factor intensity of that type, with no effect on wages. The labor supplied by and demanded of immigrant workers in agriculture may also differ from that of native workers, and anecdotal evidence indicates it is difficult to procure native workers to do many manual agricultural tasks, even for relatively high wages (Clemens, 2013; Peri and Sparber, 2009). In addition to potentially not having 'competing' workers, other factors of production, like machinery, may only partially substitute for labor.

Clemens et al. (2018) develop a theoretical framework to analyze the impact of a labor supply shock on the agricultural sector. They describe three potential outcomes for agriculture after a labor supply shock. If capital, technology, or output cannot adjust, farm wages will increase. If adjustment is possible within a 'cone of diversification', factor intensity, but not wages, will increase in response to changes in factor supply. The 'cone' is essentially an area where both automated and traditional technologies are feasible and substitution between them is possible. In

[1] An effective increase in local jail revenue was not the documented intention of the 287(g) program, but the perception of a revenue opportunity was documented by immigration scholars, including Shahani and Greene (2009) and through personal correspondence with representatives of participating jurisdictions

response to the 1960s 'bracero exclusion' that Clemens et al. (2018) analyze, or more broadly to a decline in immigrant labor supply, their model of endogenous technical change predicts no changes to wages, an increase in output that uses automatic harvest technology, and a decline in output that uses traditional harvest technology. In the absence of fully substitutable automatic technologies, or when substitution is incomplete, wages will rise and output could fall to the point where alternative land uses become more profitable. Likewise, in the early 20th century, US agricultural production may have been within the cone, with operations substituting across crops (Lafortune et al., 2015) and capital (Lafortune et al., 2019) in response to labor availability. Evidence of whether modern US agriculture would respond similarly to labor supply shocks is limited.

2.2. Immigrant labor and the farm economy

Migrants make up a large share of the farm workforce in many countries. (Charlton et al., 2021). Immigration policy and enforcement have serious implications for US farms. Approximately half of US farm workers are estimated to be "undocumented", or lack legal status to work in the US (Zahniser et al., 2012). Richards (2018) estimated that in California, a 50 percent decline in undocumented workers would lead to a 22 percent increase in farm wages. While significant mechanization of US agriculture has occurred over the past several decades, many specializations still rely on hired labor to perform complicated or delicate tasks. While labor expenses are only 17 percent of cash expenses for the US farm sector, this share approaches 40 percent for more labor-intensive specializations (Zahniser et al., 2012). There may be localized shortages of farm labor across the US: Hertz and Zahniser (2013) identified several counties with farm worker wage growth of over 40 percent where agricultural employment has fallen.

The farm sector provides a unique setting for causal identification of the firm-level impacts of a specific immigration policy. Changes in outcomes in other industries that utilize undocumented labor, such as hospitality and construction, are difficult to untangle from general economic trends. However, the farm sector is not strongly tied to general economic conditions. For example, during the recession that began in 2008, US farms were much less affected than the rest of the economy (Shane et al., 2009). Hornbeck and Keskin (2015) show that productivity, revenue, and land value gains in agriculture largely do not influence the non-agricultural sector. The economic spillovers of agricultural gains are limited to only the immediate short run and are not persistent. Weber et al. (2014) find a similar result using recent data on crop price shifts, confirming only weak links between farm revenue and nonfarm income or employment. This may reflect, to some degree, a small and stable farm population: the number of US farms has been stable at slightly over 2 million since 1970[2].

As in the general literature on labor and immigration policy, studies on farm labor issues have focused on wages, labor supply, and migration decisions (e.g., Buccola et al., 2012; Fan et al., 2015; Taylor et al., 2012). Kostandini et al. (2014) find that authorization of local (county) immigration enforcement through 287(g) in 2007 led to a decline in non-citizen population levels, based on population estimates from the American Community Survey. They also consider the association between sector-level outcomes stemming from initial 287(g) authorization in 2007. However, they do not make causal claims about how immigration enforcement impacts the agricultural sector at the firm level.

2.3. The 287(g) program

Over the last decade, localities have increasingly sought ways to signal their anti-immigrant sentiment, motivated in part by discontent with state- and national-level approaches. For example, some municipalities have attempted to pass ordinances requiring proof of legal residency to obtain housing or have designated English as the "official language" of their municipality, a non-binding and unenforceable measure (Guzman, 2010). Other jurisdictions moved from signaling to explicit action through an immigration enforcement program now widely known as 287(g). This program, outlined in Section 287(g) of the 1996 Immigration and Nationality Act, allows law enforcement officials at the state, county, or city level to be deputized as national immigration agents and enforce national immigration policy within their jurisdiction.[3] Jurisdictions elect into and are approved for one of two 287(g) "models", the jail model or the task force model, or can choose to have both. The jail model limits enforcement to checking immigration status as part of the intake process for someone already being jailed.[4] Enforcement reaches farther with the task force model: for example, local officials can check a person's immigration status during a routine traffic stop or any police encounter in the field. If the person does not have legal status, the 287(g) deputy can then begin the procedure to remove him/her from the United States.

While the 287(g) program's stated intent was to identify and remove only dangerous undocumented individuals, analysts debate the extent to which implementation aligned with this intention. Opponents have argued that it provides the means for local law enforcement to racially target and harass residents of their jurisdictions (Shahani and Greene, 2009). Although the direct removal of immigrants was nontrivial in many places, the program's true impact may have been as a signal to residents, especially Hispanics, who may be targeted by the program, regardless of their immigration, documentation, or even citizenship status. Indeed, proponents of the program highlighted this encouragement of out-migration as a benefit of the program (Capps et al., 2011). This signalling effect is by no means limited to 287(g): a program that mandated E-verify in Arizona in 2007 was shown to reduce the population share of "Hispanic non-citizens" relative to other states (Bohn et al., 2014). Through both direct action and indirect signalling, implementation of 287(g) is related to a decrease in the local immigrant population and immigrant labor, regardless of legal status (Kostandini et al., 2014; Watson, 2013).

Because there is temporal and spatial variation in the implementation of this program, it provides a unique opportunity to study the impacts of local, county-level shocks to the population of immigrants. Bohn and Santillano (2017) used a contiguous-county pairs identification strategy for 287(g) participation and find diverse county-level employment impacts, while Dee and Murphy (2020) use difference-in-differences and triple differences to find that 287(g) reduces the population of Hispanic students in a jurisdiction by 10% over two years. Many contributions to this area have focused on aggregated measures

[2] see https://www.ers.usda.gov/data-products/ag-and-food-statistics-charting-the-essentials/farming-and-farm-income/

[3] The actual language of Section 287(g) is vague with regards to the scope of the program; this lack of guidance contributed to low utilization of the program prior to the creation of the Department of Homeland Security, which was created in 2003 and took over immigration enforcement from the Justice Department. Salt Lake City, Utah was the only jurisdiction to hold formal negotiations for a 287(g) agreement in the 1990s; the agreement was never signed due to public concerns around racial profiling. Interest in the program increased after 9/11, when local jurisdictions became more invested in finding ways to participate in counter-terrorism efforts; the first official agreement was signed in 2002 with the State of Florida. There were seven agreements prior to 2006; these early agreements were motivated by concerns about states' limited ability to combat terrorism locally. The shift towards the "universal" model of enforcement and away from targeting only violent criminals occurred in 2006, as part of growing anti-immigrant rhetoric and concerns about "illegal immigration" during that time (Capps et al., 2011).

[4] In this model, the initial booking officer does not have to be associated with a 287(g) law enforcement agency; the person being arrested can be directed to a 287(g) facility where a 287(g) officer interviews them to determine their immigration status. Critics of the program frequently point to this as an area rife for racial profiling (Capps et al., 2011).

of employment or industry-level outcomes following 287(g) implementation, such as (Kostandini et al., 2014), who estimated the immediate (2007 only) association between 287(g) participation and 2007 noncitizen population and some county-level indicators of farm labor supply shocks. Here, we look at the farm sector's response to this negative labor supply shock. Our outcomes include measures of the shock's magnitude, shorter term production decisions to evaluate the possibility of endogenous technical change, and longer-term estimates of the impact on farm profitability and financial stability.

A jurisdiction's self-selection into the 287(g) program was typically a response to political pressure, and so codified an unwelcome environment for immigrants that existed prior to program participation.[5] The heads of law enforcement agencies (LEAs) interested in participation approached ICE via letter or email. ICE would then conduct a field survey to determine whether the LEA had the "capability to fully support the request;" there was no official requirement for public notice or negotiation.[6] Final approval for participation came from the agency's Assistant Secretary, after which a Memorandum of Agreement (MOA) was signed by the Assistant Secretary of ICE and the head of the LEA (Immigration and Customs Enforcement, 6 Sept. 2007).[7] Because these requests are ultimately approved or declined by ICE administration, there is a group of counties that wished to participate but could not. In Section 5.2.2, we extend our analysis to include these counties, picking up the impact of 287(g) as a proxy for local anti-immigrant sentiment, which presumably exists in all the counties that applied for 287(g) authorization, regardless of official approval. In 2012, the last year of our study, 287(g) was scaled back, to eventually be supplanted by other programs, including Secure Communities. Secure Communities began its pilot period in 2008 with 14 jurisdictions. Unlike 287(g), which also had a component not administered in jails, Secure Committee's focus is almost entirely on identifying illegal aliens either already incarcerated or at the point of arrest, i.e., when fingerprinted (Immigration and Customs Enforcement, 2018 (accessed February 3, 2020).

3. Data

3.1. Immigration enforcement data

In 2007, twenty-four (24) jurisdictions signed MOAs with ICE and enrolled in the 287(g) program. By 2010, the number had reached its peak with fifty-four (54) jurisdictions enrolled; these are pictured in Fig. 1. As the map shows, participants are concentrated in the south, although a range of agricultural production systems are represented. In our analysis, the level of treatment is at the county level, meaning that a county is considered in the program if its own law enforcement body, generally Sheriff's Offices, has 287(g) authorization (83% of our treated sample), or if city in the county has signed an approved 287(g) agreement between its police department and ICE (9% of our treated sample).[8] In our analysis, we use all 287(g) programs regardless of their implementation model (jail, task force, or both). We show in our robustness checks (Section 5.2) that our results are driven by counties that include task force.

A county signing a MOA with ICE measures participation in 287(g) on the extensive margin. In data accessible via a Freedom of Information Act (FOIA) request, ICE provides annual (and, in some cases, monthly) measures of the intensity of 287(g) enforcement for each participating jurisdiction. 'Aliens identified' measures the number of undocumented immigrants identified by local deputized officials, and 'aliens departed' measures the number of those that were successfully removed from the jurisdiction.[9] As such, aliens departed is the stronger measure of enforcement, although both measures quantify the extent to which a county acts on their 287(g) mandate. Appendix Table C1 has summary statistics for enforcement data, showing the high degree of variability of 287(g) enforcement measures across counties.

There are 140 jurisdiction-year observations where no aliens were identified (33.0% of the 424 jurisdiction-year observations), and 138 (32.5%) during which none were departed. A 287(g) program both directly reduces the supply of immigrant labor in a county, but also discourages potential immigrant laborers. Watson (2013) shows that the 287(g) task force model significantly discourages immigrant inflows to a 287(g) location, in addition to pushing immigrants from 287(g) areas. By Watson's calculations, the impact of this out-migration is equivalent to a 15% decline in predicted labor demand. Importantly, Watson's work also shows that these local immigration enforcement policies do not drive workers out of the United States entirely, or discourage undocumented immigrants from entering the US, but rather cause within-country or within-state migration between local areas that have the program and those that do not.

3.2. Farm survey data

To test whether such an outflow of immigrant labor is observed in the agricultural sector, and to estimate the impact of that outflow, we use data on production decisions and farm structure that come from two national farm surveys, both conducted by the United States Department of Agriculture (USDA). The Agricultural Resource Management Survey (ARMS) is the only nationally representative annual farm-level survey. This cross-sectional survey collects detailed data on production activities, finances, and household characteristics. In most years, approximately 20,000 farms complete ARMS. Data collected from ARMS inform official agricultural statistics and have supported a wide body of research (Kuethe and Morehart, 2012).

ARMS uses a stratified random sampling procedure. Because of its large sample size relative to the US farm population, over 60,000 farms have been included in ARMS at least twice since its inception in 1996. Weber et al. (2016) took advantage of these repeated observations to create an unbalanced panel to analyze the impact of crop insurance participation on fertilizer expenditure. We use a similar unbalanced panel to analyze the impact of 287(g) authorization on various farm production decisions. While farms are randomly selected, larger farms are over-sampled due to their low numbers relative to the rest of the population and lower response rates. While this may be an issue for studies that want to draw implications for the entire farm sector, our study is concerned with farms that are labor intensive or use hired labor. Farms in the ARMS panel tend to be relatively large, as detailed by Weber et al. (2016), and so this sampling design provides us farms that are more likely to impacted by immigration policies. Of the approximately 23,000 farms in our sample, nearly 500 are located in 287(g) counties.

[5] The process by which a jurisdiction sought a 287(g) agreement varies: commonly, law enforcement officials (typically sheriffs) seeking election promised a 287(g) agreement as part of their campaign. In some states, pressure for agreements came from immigration-control lobby groups. Typically, the greater the political pressure for a program in a jurisdiction, the greater the extent of the program in targeting civil immigration violators as well as violent criminals.

[6] In fact, Shahani and Greene (2009) record at least one instance of a jurisdiction's residents (Hudson County, NJ) not knowing there was a 287(g) agreement in place until the signed MOA appeared on the agency's website.

[7] The governor or another "senior political entity" were also authorized to sign these MOAs (Immigration and Customs Enforcement, 6 Sept. 2007); direct observation of the archived MOAs shows that the head of the LEA almost always signed.

[8] In addition to the city and county agreements described here, 13 states also had 287(g) agreements, typically between ICE and the state's Department of Corrections. Because these agreements generally only impact already incarcerated immigrants, their effect on farm workers is likely to be low.

[9] We follow the official language used by ICE here: hence the use of "departed" to describe this action rather than the more commonly seen "deported".

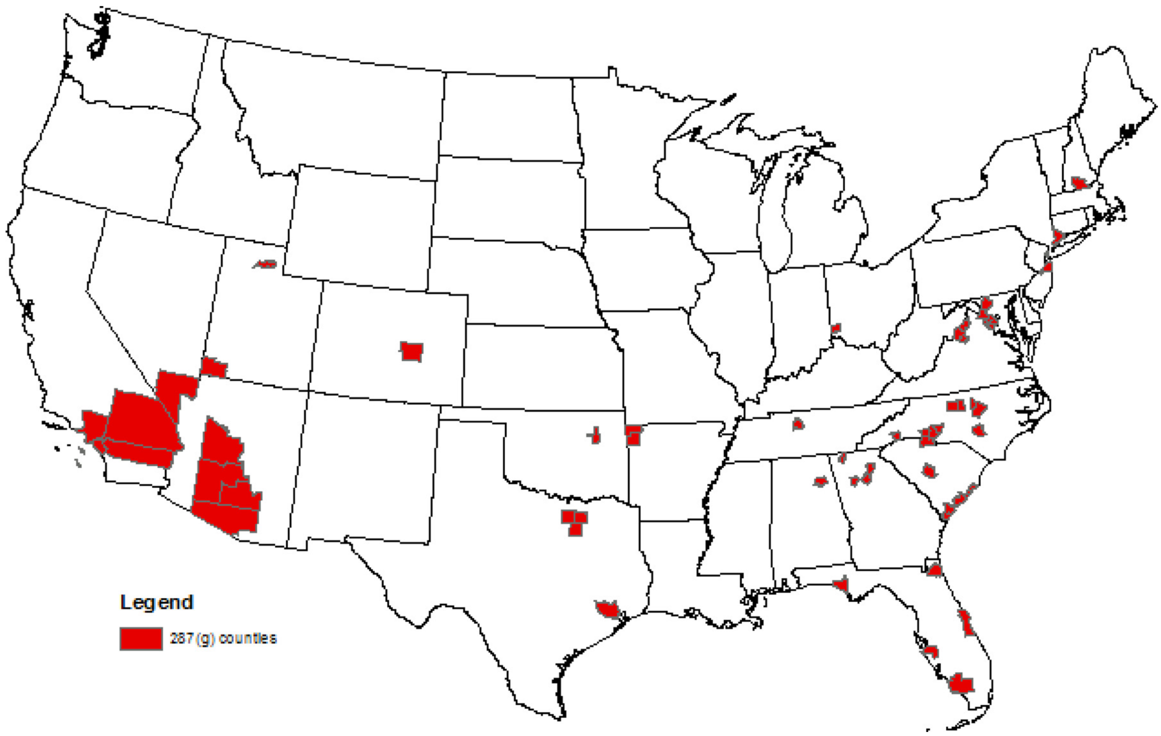

Fig. 1. Counties with a 287(g) program.

Table 1
Summary statistics: Farm production variables, pre-287(g).

	All counties			287(g) counties			Non-287(g) counties		
	mean	sd	n	mean	sd	n	mean	sd	n
Acres operated/farm[a]	1670.6	8327.7	11,730	1146.9	7850.9	439	1,691.0	8,345.3	11,291
Acres operated / county[b]	106,550.3	176,354.9	1288	98,424.8	163,268.0	52	106,892.2	176,937.5	1236
Fuel expenses ($)[a]	$ 33,879.4	$ 82,955.0	11,730	36589.8	$ 106,976.0	439	$ 33,774.0	$ 81,883.1	11,291
Fuel expenses ($)[b]	$ 2,077,636	$ 3,937,702	1320	$ 2,899,264	$ 3,758,092	53	$ 2,043,267	$ 3,942,725	1267
Labor expenses ($)[a]	$ 169,791.3	$ 885,674.1	11,730	234,717.9	$ 749,999.6	439	$ 167,266.9	$ 890,464.7	11,291
Labor expenses ($)[b]	$ 39,466,877	$ 92,217,535	1242	$71,449,801*	$ 130,710,505	50	$ 38,125,312	$ 90,074,548	1192
# of workers / county[b]	1,089.8	3,310.8	1324	2,256.4**	3,982.4	53	1,041.2	3,272.6	1271

[a] Outcome from ARMS
[b] Outcome from Census

3.3. County-level farm survey data

While survey weights are provided for ARMS to calculate population-level statistics, they are only applicable to single-year analyses due to survey design. Hence, our estimates will be representative of farms that were randomly sampled more than once over our study period (1996–2012). While the firm-level panel structure of this data set is an advantage, farm attrition is a potential concern, as is the external validity of our sample. As ARMS represents farms that were in operation before and after 287(g) authorization, we complement our ARMS analysis with publicly available county-level Census of Agriculture data, which are collected from all US farms every five years. Response to the Census of Agriculture is mandatory and so represents a much broader population than ARMS; these data allow us to understand how labor supply shocks affected the entire farm economy. Census data provide an opportunity to understand the local farm sector and individual farm impacts. In order to standardize these measures, all Census outcomes are normalized relative to the number of agricultural acres in the most recent Census year prior to 287(g) authorization. This process addresses concerns that larger (or more agricultural) counties are driving the results, and provides for a degree of comparability of the estimated coefficients across counties and across the dependent variables. The county-level Census variables are *not* replications of information reported from ARMS, because variable availability differs, and, more importantly, the data sets each represent a different population of farms. However, both data sets provide complementary information on endogenous technical change in agriculture after implementation of 287(g).

3.4. Dependent variables

We use the theoretical model of Clemens et al. (2018) to inform our choice of outcome variables from available data reported in ARMS and the Census. Under this framework, we consider the impact of 287(g) authorization on labor expenditure, fuel expenditure, number of farm workers, and acres operated. In our selection of outcome variables, we complement Clemens et al. (2018)'s primary focus on state-level wages, employment, and harvested acreage by crop to estimate how other key indicators of endogenous technical change respond to local changes in immigration enforcement. We are unable to directly replicate their dependent variables due to data availability constraints, particularly around wages. When possible, we complement the farm-level outcome from ARMS with a county-level measure from the Census and vice versa. Table 1 reports average values of these outcomes prior to program onset.

By analyzing the impact on number of workers, we are able to confirm the finding of Kostandini et al. (2014) and others of a 287(g)-induced negative labor supply shock. Endogenous technical change predicts no change to wages within in the 'cone of diversification'. Although hourly wage data is not available, labor expenditure is reported in ARMS (per operation) and the Census (per county), and the number of workers per county is reported in the Census. Although under endogenous technical change no increase in wages is possible, labor expenditure could decrease, if, for example, the number of hours worked declines with higher use of automatic technology. On the other hand, if there are more dynamic changes due to operators' inability to substitute for workers or technology, the impact on labor expenditure is indeterminate. More skilled workers may require higher wages even if hours decrease.

To test whether the use of 'automatic technologies' increases, we consider fuel and oil expenditure[10], which is available in both surveys. Given that changes to fuel prices are largely driven by global oil prices, an increase in fuel expenditure would suggest an increase in use of automatic technologies that require more fuel. While we cannot precisely measure acres operated with automatic technologies, we can measure total acres operated by operation (ARMS) and by county (Census). A decline in acres would suggest that full substitution from one production type to another was not possible, and that alternative land uses became more profitable.

4. Empirical model

Our empirical model tests for evidence of endogenous technical change in response to a local labor supply shock. The outcome variables described above are modeled as a function of 287(g) program authorization (effectively, codified immigration discouragement), which varies spatially and temporally. We restrict our analysis to states that contain at least one local 287(g) program in our main results. This limits the extent to which our results could be driven by comparing outcomes across very different agricultural production areas and increases the comparability between program and non-program counties. Our estimating equations take into account the different start dates for each jurisdiction's 287(g) program; our measure of program participation is an indicator variable equal to one if the jurisdiction had a signed MOA in that year.[11] We also control for whether or not a non-participating county bordered a 287(g) county, because there could be spillovers to neighboring counties, as suggested by Watson (2013). It is uncertain *a priori* whether 287(g) would increase or decrease immigrant labor supply in nearby counties without the program. The 287(g) program may have deterred immigrants from an entire area; immigrants may also have relocated from a program county to neighboring counties, which resemble program counties in many dimensions but lack the explicit local immigration enforcement. Similar estimates of program impact for 287(g)-participant counties and 287(g)-border counties could suggest general economic or property value trends as a potential confounder, or at least motivate further investigation of these trends.

4.1. OLS model

We use a farm fixed-effects model with the ARMS data, as time-invariant farm-level characteristics, such as specialization, have a strong relationship with labor use. While our ARMS panel is unbalanced, we expect no bias other than that imposed by (previously discussed) stratification, as farms are observed based on random selection into ARMS. However, farms had to be sampled after the policy went into effect, so our results are interpreted as representative of farms that were in operation before and after the policy. Our basic estimating equation is as follows:

$$Y_{ict} = \alpha_0 + \alpha_1 G_{ct} + \alpha_2 B_{ct} + \tau_t + \gamma_i + \varepsilon_{ict} \tag{1}$$

where Y_{ict} is the outcome of interest for farm i in county c at time t; G_{ct} is either an indicator for 287(g) participation or a measure of 287(g) enforcement in county c and year t; B_{ct} is an indicator for whether a non-287(g) county c borders an active 287(g) in year t, τ_t are year fixed-effects, γ_i are farm fixed-effects, and ε_{ict} represents variation in the dependent variable that cannot be explained by the model. For our analysis of ARMS data, we cluster standard errors to be robust to correlation at the strata level[12] to take into account survey design, based on the recommendation of Weber et al. (2016).[13]

Correspondingly, our estimating equation for Census of Agriculture data is as follows, with the unit of observation being a county c and year t, with $t \in [1997, 2002, 2007, 2012]$:

$$\frac{Y_{ct}}{A_{c\bar{t}}} = \alpha_0 + \alpha_1 G_{ct} + \alpha_2 B_{ct} + \tau_t + \gamma_c + \varepsilon_{ct} \tag{2}$$

As discussed above, outcomes from the Census of Agriculture are normalized by the acres operated in the county (A_c) in the most recent Census year prior to program authorization, denoted $\bar{t}$. Results from the estimation of these naive models are reported in Table 2.

4.2. Instrumental variable strategy

The farm or county fixed-effects regressions described by Eqs. 1 and 2 will not provide causal estimates of the effect of 287(g) participation if participation is correlated with other economic factors that also influence farm sector outcomes. Farms located in 287(g) counties are different in many dimensions than farms in other counties (see appendix Table A.2), and these counties do have a higher number of pre-program farm workers (Table 1). Changes in broad farm-sector trends appear be similar between 287(g), border and other counties, and key population trends are also similar (appendix Table C2). However, participation in 287(g) itself is not random, as law enforcement agencies select into the program. Participation and the presence of immigrants in a jurisdiction could be subject to reverse causality: a high immigrant population could make 287(g) more popular, while at the same time enforcement decreases the immigrant population directly through removals and indirectly by signalling hostility towards immigrants. While our use of farm (county) fixed-effects controls for time-invariant farm (county) characteristics, including static political and geographic conditions, we cannot prove that broader social or economic factors are not driving farm decisions, especially around labor use, and 287(g) participation or enforcement levels. We thus address the potential endogeneity of 287(g) participation by using measures of local law enforcement intensity before 287(g) program authorization as instrumental variables (IV).

As part of our preliminary research on the 287(g) program, we spoke to representatives from about one-third of enforcement units that had ever been enrolled in 287(g) about their interest in the program. The most frequently shared for motivation for 287(g) participation were first, "politics" and second, the belief that there was the potential to earn revenue through the program through reimbursements for housing suspected unauthorized immigrants. While we found insufficient variation in voting patterns for an 'election outcomes' instrumental variable, various measures of local capacity for housing detainees are highly correlated with 287(g) authorization. We discuss potential confounding ef-

[10] This measure includes fuel (i.e., diesel, gasoline, etc.) and oil but excludes any utilities or electricity. The vast majority of farm machinery and equipment is powered by fuel.

[11] Three jurisdictions withdrew from the program: two in 2009 and one in 2010. These jurisdictions are only recorded as treated in the years their MOAs were active.

[12] Strata are based on value of farm sales, but the strata variable is not available before 2005 in our data set. We group estimated value of farm sales into 30 clusters, based on pre-287(g) gross cash farm income and value of contract production.

[13] We discuss the implications of different levels of geographic clustering for our standard errors, and report our main results estimated with these other formulations of the standard errors in C.4.

Table 2
Impact of 287(g) authorization on production decisions: OLS.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
287(g) authorization	–473.3	–0.156***	3110	3.72	9325	12.9	–0.004
	(347.5)	(0.043)	(9,088)	(6.43)	(18,438)	(18.3)	(0.003)
287(g) border	–100.2	-0.062**	2061	11.3	–15,576	38.8**	0.006
	(81.63)	(0.028)	(3,978)	(8.49)	(26,006)	(17.2)	(0.005)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	21,839	4889	21,839	4939	21,839	4705	3691
Number of counties	-	1240	-	1240	-	1238	1238
Number of farms	11,753	-	11,753	-	11,753	-	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

fects for this relationship in Section 4.2.3, and see only limited viability for any alternative explanations.

4.2.1. County jail activity and the 287(g) program

In addition to offering local law enforcement a "political trophy in local law enforcement campaigns", law enforcement agencies may have perceived the potential for financial gains from the 287(g) program (Shahani and Greene, 2009). Despite statutes disallowing such reimbursements, there have been concerns that ICE "misrepresented" the extent to which reimbursement was possible. Shahani and Greene (2009) cite evidence that, even if this was not actually true, local law enforcement were under the impression that they would be reimbursed for the cost of housing incarcerated non-citizens under the 287(g) program. In addition, physical capacity constraints in local jails meant that a jurisdiction with more space in its local jail was more readily able to participate than a jurisdiction whose incarcerated population was near its jail's rated capacity. Therefore, we use changes in a county's aggregated jail occupancy[14] over time, which measures the number of "empty beds" in county jails, prior to authorization, to instrument for program participation.

Data on jail occupancy come from the Census of Jails, which is conducted by the Bureau of Justice Statistics every five to six years and surveys every locally operated jail facility in the United States (Bureau of Justice Statistics, 2007; Bureau of Justice Statistics, 2010).[15] Measures are taken to ensure that complete coverage is achieved for a small subset of variables, including the jail's rated capacity (the total number of inmates the facility can legally hold) and the total population of inmates. To construct our instrument, facility-level measures are aggregated up to the county level; values from the 2005/2006 Census are used to measure pre-program capacity for 2007 and later, and values from the 1999 Census are used before 2007.[16] We define occupancy as the total population subtracted from the rated capacity, such that a negative value for occupancy indicates prison overcrowding and a positive value indicates an emptier facility.

Appendix Fig. C1 shows the distribution of jail occupancy across the United States. Although there are certainly areas where jail overcrowding occurs, there is no evidence that there is meaningful spatial autocorrelation. The Moran's I, a standard measure of spatial correlation, is not significantly different than zero using the minimum threshold distance spatial weights matrix for US counties (see appendix Fig. C2). Although this does not definitively prove that jail occupancy in one county does not impact occupancy in its neighboring counties, it alleviates concerns that county-level jail populations move together within a region.

4.2.2. 2SLS model

All of the coefficients in our main results are estimates of the local average treatment effect (LATE). We estimate the effect of program participation only for compliers, or the sub-population of counties whose 287(g) status was manipulated by jail occupancy. The counties who would have participated in 287(g) regardless of their jail occupancy are not part of this group: our instrumental variables strategy identifies off of counties with a more active local law enforcement culture or counties with higher investment in law enforcement infrastructure relative to incarnation rates or crime levels. We first report reduced form estimates with jail occupancy as the independent variable of interest on the set of dependent variables described in Section 3.4. These results are presented in appendix Table C5. Ultimately, we are interested in how farms adapt to labor shortages driven by changes in immigration policy enforcement. To the degree that our estimates capture local enforcement relative to actual 287(g) participation, our estimates should still reflect the impact of enhanced enforcement on firm behavior.

Our estimating equation using jail occupancy levels as an instrumental variable ($Z_{ct)}$) for 287(g) participation is as follows, with instrumented 287(g) participation represented by G^*_{ct} and the other variables defined as in Eq. 1:

$$Y_{ict} = \alpha_0 + \alpha_1 \underbrace{G^*_{ct}}_{=Z_{ct}} + \alpha_2 B_{ct} + \tau_t + \gamma_i + \epsilon_{ict} \tag{3}$$

We also estimate an equivalent specification, with county-year observations and county fixed effects for use with Census of Agriculture data.

$$\frac{Y_{ct}}{A_{c\bar{t}}} = \alpha_0 + \alpha_1 \underbrace{G^*_{ct}}_{=Z_{ct}} + \alpha_2 B_{ct} + \tau_t + \gamma_c + \epsilon_{ct} \tag{4}$$

4.2.3. Exclusion restriction

The validity of our jail occupancy instrumental variable rests on the plausibility of local jail capacity affecting farm decisions and outcomes *only* through its relationship with 287(g) program participation. It is

[14] Vacancy may be a more accurate term in this case, but we use 'occupancy', as it is less emotive and a more general term that can encompass either overcrowding or unused space.

[15] States with combined jail-prison systems (Connecticut, Delaware, Hawaii, Rhode Island, Vermont, and Alaska) are not included in the Census. Only one 287(g) jurisdiction (City of Danbury, CT) is in one of these states and so is dropped from our analysis, and none are states with a particularly large agricultural sector.

[16] The Census of Jails was scheduled to occur in 2005; however, it was split into the 2005 Census of Jail Inmates and the 2006 Census of Jail Facilities in order to ease administrative burden. Each iteration reported total population and rated capacity with almost universal coverage.

therefore necessary to rule out potential factors, omitted from our main specification, that may be able to explain the relationships we observe. In Appendix A, we consider a variety of factors and trends, including a county's economic specialization, weather, and crime, that may be related to both farm outcomes and local law enforcement and find no evidence to suggest that there is any consistent relationship between them. In particular, we address the possibility that urbanization or population growth (either increasing or decreasing) is the omitted factor driving these results. Further, the Section 5.2.3 robustness checks show our results are largely robust to excluding urban counties and are driven by county 287(g) programs rather than those in cities. Ultimately, we find no factor that explains how program counties, which are spread across the country, differ structurally from their neighboring counties in a way that would provide a causal pathway from jail occupancy to farm or agricultural sector outcomes. Because we examine agricultural outcomes in the relatively few counties with 287(g) programs, the non-program counties in those states, including counties that border 287(g) counties, form the control. Any economy-wide (or even state-wide) trends that could drive our findings would have to be localized in program counties. Indeed, any such effect would have to impact these counties differently from their neighbors, but have the same effect across the 287(g) program group, which represents a variety of economic, agricultural, and even legal systems, to the extent that laws and regulations differ by state and county. We continue and expand this argument in our discussion of the main results (Section 5.1) and test its validity in with a variety of robustness checks (Section 5.2).

5. Results

In this section, we report results on the impact of county 287(g) participation on farm- and county-level measures of endogenous technical change. We first estimate Eqs. 1 and 2: the impact of 287(g) at the farm (ARMS) and county (Census) levels without an instrumental variable and report the results in Table 2.[17] These OLS estimates are average treatment effects assuming that program participation G_{ct} is independent of the error term ε_{ct}; this assumption is unlikely to hold, as 287(g) participation is endogenous through reverse causality or through omission of county- or farm-specific trends correlated with both program participation and farm outcomes. Participation was non-random: OLS estimates are more than likely biased by the counties' self-selection into a 287(g) agreement.

Therefore, the OLS estimates are systematically much lower than the results from estimating Eqs. 3 (farm-level) and 4 (county-level) using 2SLS. The first stage results are reported in appendix Table C6. The first stage confirms that jail occupancy satisfies the inclusion restriction and is strongly correlated with 287(g) participation using the standard benchmark. The 2SLS coefficients represent the local average treatment effect on the set of compliers in our sample. The compliers are the jurisdictions induced to enroll in 287(g) as a result of increasing jail capacity in their local area. Following Angrist and Pischke (2008), we estimate the set of compliers in our sample to be 60% of the program counties. Although it is not possible to perfectly observe the compliers, it is reasonable to assume they would be more zealous participants in the program than the average county in our sample, as the difference between the likely biased and much lower ATE estimates in Table 2 and the LATE estimates in Table 3 would indicate.

Because these agreements tended to be the result of community pressure or promised as part of a political campaign for elected law enforcement officials, the start of an agreement in an individual jurisdiction was not necessarily a shock. We cannot rule out anticipatory behavior in the pre-treatment period from farm operators or farm workers. Therefore, the estimates presented below underestimate the true impact of a 287(g) agreement.

Following these results, we present a variety of robustness checks, starting with an exploration of a potential mechanism driving jail occupancy in program counties (Section 5.2.1). Then, we test whether the main results are sensitive to various changes in the model specification, including alternative measures of enforcement (Section 5.2.2) and different compositions of the treated and control groups (Section 5.2.3). In 5.2.3, we also describe results from placebo tests designed to eliminate concern that either time trends or idiosyncrasies about these particular counties drive our results. In our final results section, we summarize key findings of a supplemental analysis of 287(g)'s impact on specific acreage allocations and on the financial health of the local farm economy and the financial viability of individual farms in program counties.

5.1. Main results

In Table 3, we report two coefficients from our main specifications, described by Eq. 3 (farm-level) and Eq. 4 (county-level). The coefficient α_1 describes the impact of program participation in a program county and α_2 describes the impact on a county that borders at least one program county. All of these spillover results are in the same direction as in program counties, albeit at smaller magnitudes. This provides further evidence that the program's effect on the labor force was through signaling a hostile environment for immigrants, in addition to direct action. Unlike the enforcement action, the hostility and negative sentiment were not confined to jurisdictional borders. Fig. 1 shows there are definitive clumps of program counties: the tendency for a county to enroll in the program if one of its neighbors did was quite high. The spillover effects we find could also reflect immigrants in those places responding preemptively to that tendency.

The effects of 287(g) program participation reported in Table 3 are consistent with a negative labor supply shock. Fuel expenses per farm and per county both increase as a result of 287(g) program authorization, while the number of workers and number of acres operated in a county both decline. Labor expenditure increased at the county level; labor became more expensive, either through higher wages or a change in the workforce composition, such as the use of more managerial or technical labor (i.e., mechanics and equipment operators). The decline in the number of workers combined with the increase in fuel suggests farms' attempts at substitution, while the decline in acres is evidence that these efforts fell short of maintaining previous levels of production. The increase in labor expenditure further suggests non-marginal changes to production technology outside of the 'cone of diversification' discussed by Clemens et al. (2018), as endogenous technical change would imply no increase in wages.

We find that the primary change at the farm level was in fuel expenses, which increased substantially for farms in program counties. Increases in fuel expenses show that farm operators increased their use of equipment, and suggest that farm operators had to replace farm labor with equipment. Fuel-powered machinery is one of the only available substitutes for farm labor, although its effectiveness may not extend to all tasks. Our coefficient suggests an average increase of nearly $81,000 per operation in 287(g) counties. Fuel is a major farm expense, averaging nearly $37,000 per farm in a 287(g) county. This large coefficient suggests relatively larger farms increased fuel use in response to 287(g). We see a corresponding increase in county-level fuel expenses, with program counties experiencing an increase in fuel expenses of more than $200 per pre-program-acre-operated in the county. Given the magnitude of the effect and the lack of a significant spillover effect, this result is likely driven by using more fuel, rather than by increases in fuel prices.

The most direct evidence of a negative labor supply shock comes from column (7) of Table 3: the number of agricultural workers in 287(g) counties declined by 0.08 workers per pre-program-acre-operated. Given the size of this estimate and that the average pre-program acreage was about 80,000, counties with more labor intensive

[17] The reduced form estimates of the impact of jail occupancy itself on the dependent variables are available in appendix Table C5.

Table 3
Impact of 287(g) authorization on production decisions: 2SLS.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
287(g) authorization	–446.7	-3.67***	80,960**	231*	515,971	686**	-0.081**
	(488.7)	(1.21)	(36,350)	(127)	(476,880)	(278)	(0.035)
287(g) border	–83.11	-0.189***	7011	19.3**	16,336	62.9***	0.003
	(56.79)	(0.051)	(5,373)	(8.4)	(13,015)	(17.2)	(0.005)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	21,403	4874	21,403	4923	21,403	4690	3681
Number of counties	-	1239	-	1239	-	1237	1237
Number of farms	11,501	-	11,501	-	11,501	-	-

[a] Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses [b] Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. [c] *** p<0.01, ** p<0.05, * p<0.1 [d] Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

agriculture were likely most affected by 287(g). It is unlikely, however, that a decline of this magnitude is the result of direct 287(g) action; instead, it is driven by the signalling impact of 287(g) for immigrants, both with and without legal authorization to work.

In addition to the impacts on the size of the labor force, the area operated also declined following 287(g) implementation. Program counties operated significantly fewer acres after 287(g) authorization, although the farms in the ARMS sample are not decreasing the size of their individual operations. Counties with 287(g) programs saw a contraction in their agricultural land; acres operated in treated counties declined by 3.67%, a loss of over 3000 agricultural acres from the average program county. Loss of farmland at the county rather than farm level is not surprising, given the composition of the two data sets and the general illiquidity of farmland (Barry et al., 1981). The farm-level sample is more representative of large, commercial-scale operations. Although we see no effect of program participation on the size of these operations, results using the Census data (see Appendix B.2, Table B4) show a significantly smaller number of farms in program counties. While detailed analysis of structural change in the farm sector is beyond the scope of this study, the difference in results using farm- versus county-level data suggest that larger farms may have been better able to survive in a higher labor cost environment.

Finally, we find a statistically significant increase of $686 per pre-program-acre-operated in labor expenses at the county level, but no impact at the farm level. This increase belies numerous changes to labor use, including labor type, wages and level of farm production.[18] While wages likely increased given the decline in number of farm workers, we cannot disprove that farms switched to more skilled or managerial labor, rather than a relative increase in wages for manual labor. The observed increases in fuel expenses and a decline in acreage provide evidence for some substitution between worker types, such as those that could operate mechanical harvesters rather than those who harvest crops manually.

Overall, our results suggest that some farms, particularly the larger ones better represented in the farm-level sample, were able to adapt to the 287(g)-driven labor shock, but the decline in acreage suggests only limited endogenous technical change in response to the program. More broadly, farms appear not to be operating within the 'cone of diversification' described by Clemens et al. (2018). In other words, traditional technologies typically cannot be perfectly substituted for automated technologies, so dynamic changes to farm structure may occur in response to persistent changes to immigration enforcement or other declines in migrant labor.

5.2. Robustness checks

We conduct multiple robustness checks for our main specification. In addition to being generally robust to different approaches for standard error estimation (see C.4), our results are largely consistent under different measures of jail occupancy, noisier measures of 287(g) implementation, and a variety of control groups. We also conduct a series of placebo tests to test the validity of our instrumental variable strategy.

5.2.1. Drivers of jail occupancy

The Census of Jails includes information on the construction and renovation of jail facilities within a jurisdiction; these data indicate that 287(g) counties were making larger investments in local jail capacity well before the program was authorized (appendix Fig. C3). Appendix Table C3 contains statistics on jail renovation and construction by 287(g) status: in almost every pre-287(g) year, future 287(g) counties were constructing or renovating their jails. Jails in 287(g) counties are newer, with the average year of construction being 1987, versus 1980 for counties that are neither program nor border counties. Appendix Fig. C4 shows total renovations between 1997 and 2006 by county across the US.[19] High levels of law enforcement activity or a more 'active' local law enforcement culture are likely key drivers of jail renovations and interest in 287(g) participation. Even if local population growth or crime was a factor behind each of these, the lower levels of occupancy in 287(g) counties suggest disproportionate investment in jail capacity, and the differences in jail renovation behavior between 287(g) and non-287(g) counties point to a potential mechanism that explains why the jail capacity in a county is predictive of its interest in the program.

Construction of new jails and renovation of existing ones in the decade prior to 287(g) authorization could be one mechanism through which counties expanded jail capacity, which in turn influenced program participation. To test this, we use the running total of jail renovations in a county as an alternative instrument and report the results in appendix Table C7. Although magnitudes and significance levels are lower, the results do show similar evidence of a labor supply shock: increasing fuel expenses at both the farm and county level, as well as a direct decline on the number of workers in a county.[20] Although impor-

[18] Agricultural wage data at the county level is not available, either in the sources we use here or elsewhere. Further, disentangling these effects would require a measure of labor composition, which is also not recorded in our farm- or county-level surveys.

[19] We do not calculate spatial correlations due to the preponderance of zeros in this value.

[20] We also use both the occupancy and renovations instrumental variables together, again with consistent results. These results appear in appendix Table C8.

tant for understanding the underlying conditions that influenced program participation, we do not prefer this IV for estimation, as it relies on a smaller sample of counties than the occupancy IV, due to non-response for the renovation and construction questions.[21]

5.2.2. Alternative measures of enforcement

Magnitude of enforcement

Although we see an impact on farm production decisions using authorization alone, which suggests strong signalling effects, we also observe the intensive margin of program participation by each program jurisdiction. For each year of the program, ICE reported statistics on the number of "aliens identified" or "departed" in a given jurisdiction. "Aliens identified" is the number of potentially unauthorized individuals taken into custody by local authorities, and "aliens departed" is the number of undocumented individuals who were removed from the jurisdiction as a direct result of identification via 287(g) action.[22] We estimate our main specification using these measures of the magnitude of 287(g) participation in place of 287(g) authorization and report the results in appendix Table C9. While the coefficients are difficult to interpret in light of the issues discussed in footnote [22], this approach allows us to further validate whether we are observing a 287(g)-induced labor supply shock. Results from this specification are consistent with a situation in which counties with higher levels of implementation or enforcement were less attractive to farm workers than counties with minimal enforcement; in addition, the impact of an additional alien departed is greater than the impact of an additional alien identified across all outcomes.

Task force model only

Just as we expect counties with more enforcement to be more affected by the program, we also would expect the "task force" 287(g) model to have more of an impact on farm workers, especially because the limited evidence linking farm workers and crime (see 4.2.3) reduces the salience of the "jail" model. Counties with the "jail" model primarily identify undocumented immigrants who are already incarcerated, while the scope of the "task force" model is much more broad. There is likely to be a signaling effect regardless of program model, and so our main results include all programs. However, the task force model is a more relevant deterrent for farm workers, and so we run a specification keeping only the task force jurisdictions. These results, available in appendix Table C10, show that our county-level findings are indeed driven by counties using the task force model rather than the jail model.

Counties that applied for and were denied 287(g)

Anti-immigrant hostility or increased policing of immigrant communities is not necessarily restricted to communities with 287(g) programs. There are counties or cities that applied for 287(g) program authorization but were denied. These counties may have had similarly hostile environments towards immigrants, or similar reasons for pursuing the program, as did authorized counties, but were never officially mandated to act.[23] To test the impact of wanting 287(g), regardless of approval, we run our main specification with an indicator for application to the 287(g) program as our measure of participation. Our results (Table 4) are robust to this specification, although with lower levels of statistical significance and lower magnitudes. Counties that wanted 287(g) programs have fewer agricultural workers in the period after their application, as well as increased labor and fuel expenses. They also have fewer agricultural acres operated, as in the main specification. At the farm level, we continue to find a significant increase in fuel expenses. Taken together, these outcomes indicate declining levels of production, and support the hypothesis that the program affected farm workers' behavior by signalling an area's hostility towards immigrants. The local sentiment around immigrants, or the attitude of local law enforcement, act as a significant deterrent for farm workers with or without an official mandate to act on such feelings.

5.2.3. Alternative control groups

Without very urban counties

Concerns that growing, rather than shrinking, populations drive both jail occupancy and agricultural outcomes are mediated first by the calculation of the occupancy instrument itself. While both a county jail's population and its rated capacity are strongly and positively correlated with the total county population ($r = 0.88$ and $r = 0.84$, respectively), the difference between the two, or the county jail occupancy in our specifications, is only weakly negatively correlated with total county population in a given year ($r = -0.09$).

The counties that participated in 287(g) include some major urban areas: for example, Davidson County, TN (Nashville); Fulton County, GA (Atlanta); and Los Angeles County, CA (Los Angeles). Because these counties are atypical in terms of both population (incarcerated or otherwise) and farm outcomes, we test whether our main results are consistent when the largest urban areas are excluded from our sample of both treated and control counties.[24] Using rural-urban continuum codes (RUCC), we drop the largest urban areas from the sample, and present these results in Table 5. Our main findings from Section 5.1 on county-level acres operated, labor expenses, and number of workers are robust to this restricted specifications. The results on fuel expenses (farm- or county-level) do not hold; increased fuel costs may have been correlated with production in more urbanized counties. Many very urban counties still have large agricultural sectors, especially in the larger counties characteristic of western states. For example, Riverside and San Bernardino counties in California are defined as very urban counties and therefore dropped from this specification, but had over 300,000 and 70,000 farmland acres, respectively, in 2012.

As a result, when we drop urban areas or city-program counties (described below) from ARMS, we lose nearly half of our treated farms. Given that treated ARMS farms may or may not use hired labor and represent diverse farm specializations, generally it is not surprising that our result for fuel expenditure becomes inconclusive. However, pressure from metropolitan areas does not explain the decline in acres operated or in the number of agricultural workers, which suggests our empirical

[21] Although the 2006 Census of Jail Facilities was mandatory, approximately 33% of counties' jail(s) did not answer the jail renovation or construction questions: non-response was 18% for 287(g) counties and 35% for border counties. Appendix Table C4 compares the pre-program dependent variables across counties that did and did not report jail renovation data. Counties with jail data tend to have a more robust agricultural sector: they have more farms, and larger farms, on average, as well as more farm workers. This means that counties with jail renovation data are more likely to be responsive to any potential labor shocks. Further analysis shows that these differences are driven by the control counties, rather than the program or border counties. That is, our sample of control counties for this specification is more likely to be sensitive to a fluctuating agricultural labor supply.

[22] These measures are imperfect because they do not account for county size or the local undocumented population; however, measures of local undocumented populations are also problematic. For example, "share of noncitizens" is the best available indicator of the total population of undocumented immigrants in a county, but does not provide an accurate count of undocumented immigrants for comparison to the population "identified" under 287(g). Further, previous work has provided evidence that 287(g) reduced the population of all immigrants in a county, regardless of their documentation status (Kostandini et al., 2014).

[23] No systematic review of 287(g) program denials has been conducted, and anecdotal evidence is correspondingly slim. However, recent research by Bellows (2019) has shown that counties denied 287(g) were approved for Secure Communities sooner than other counties (at a time beyond the time frame of this study). At a minimum, this indicates that an unwillingness to engage with ICE through anti-immigration programs was likely not the reason for these counties' 287(g) program denial.

[24] Because county population may be endogenous due to the inclusion of noncitizens, we do not estimate our primary model with annual county population in the estimating equation.

Table 4
Impact of 287(g) program application (denied and accepted) on production decisions: 2SLS.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
Wanted 287(g) authorization	–456.8	–0.927**	76,839**	121**	510,813	399***	-.0454***
	(496.8)	(0.389)	(34,254)	(60)	(469,414)	(112)	(0.0129)
Wanted 287(g) border	–22.97	–0.0186	9,213*	–1.63	25,740	23.6	0.00472**
	(51.93)	(0.0309)	(5,576)	(5.65)	(23,574)	(16)	(0.00219)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	21,403	8560	21,403	8667	21,403	8253	6472
Number of counties	-	2178	-	2178	-	2170	2176
Number of farms	11,501	-	11,501	-	11,501	-	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table 5
Impact of 287(g) authorization: 2SLS without very urban counties.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
287(g) authorization	-1,095	-5.57**	-45,723	366	225,490	1,377*	-.111*
	(2,248)	(2.64)	(35,257)	(253)	(528,499)	(738)	(.0653)
287(g) border	-136.6**	-.0792	241.3	22.2*	2512	65.5**	0.00565
	(66.26)	(.0608)	(3,769)	(12.8)	(17,233)	(27.5)	(.00748)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	18,604	4002	18,604	4041	18,604	3855	3023
Number of counties	-	1016	-	1016	-	1015	1015
Number of farms	10,004	-	10,004	-	10,004	-	-

a Rural-Urban continuum code (RUCC) developed by the Economic Research Service (ERS) of USDA; very urban counties (RUCC = 1) excluded b Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses c Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. d *** p<0.01, ** p<0.05, * p<0.1 e Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

strategy isolates the 287(g) labor supply shock. Generally, this analysis suggests some interesting heterogeneity in 287(g) response which is difficult to isolate with our data sets and empirical strategy.

Without city programs

Both county and city governments had the opportunity to enroll in 287(g). During our sample period, 83% of 287(g) programs operated at the county level; we expect these county-wide programs to have more of an impact on agricultural outcomes, as they cover the rural and agricultural parts of a county where law enforcement officials are more likely to encounter farm workers. However, if our results were driven by growth in cities competing with agriculture, we would see the opposite–limited impact of the county-level programs. Our results from a specification that excludes the nine counties that contain the nine cities with a city-level programs, available in appendix Table C11, show that changes to our dependent variables from the Census are driven by county and not city 287(g) programs. These results, along with a specification controlling for a county's degree of urbanization (see appendix Table C12), address concerns that major demographic shifts drive both jail occupancy and farm outcomes. Our results are also robust to controlling for other indicators of a county's economic performance that are likely correlated with its jail occupancy rate: in appendix Table C13, we run our main specification with additional controls for the county's crime rate (disaggregated into violent crime and property crime) and the unemployment rate and find similar results.

Full control group and falsification tests

Our main results use only the non-program counties in states with at least one 287(g) program. We test the sensitivity of our results to using all counties in the contiguous United States in the control instead and find similar levels of significance, though smaller magnitudes, from this specification (see appendix Table C15). Another concern is that our results may reflect characteristics of these particular counties or nationwide trends in agriculture rather than the 287(g) program, and so we perform a series of placebo falsification tests for our Census outcomes.[25] These results do not provide definitive evidence of the validity of our instrument, but instead address concerns about pre-trends or other confounding factors driving our results in the treated counties. In the first placebo test, we naively treat the year 2002 as the year 287(g) authorizations began for the treated counties and then repeat the analysis described above, using jail capacity prior to 2002 as an instrument for program "participation" in the falsely treated year. In doing so, we effectively shift our analysis five years into the past: for example, a county that signed an MOA in 2008 is modeled as having signed it in 2003. In this test, significant coefficients would suggest that endemic characteristics of these treated counties, other than the 287(g) program, are driving our results. We find only one significant coefficient (on fuel expenses) in this analysis, which is reported in Table 6.[26]

[25] Data structure, as well as slow processing times through secure data access systems for the ARMS data, preclude us from implementing similar approaches with farm-level data.

Table 6
Census placebo test: Authorization moved 5 years earlier, 1997–2007.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Labor expenses ($) (Census)	(4) Number of farms per county (Census)
287(g) authorization	−0.0332	84.1**	1091	0.00236
	(.322)	(42.5)	(704)	(.00358)
287(g) border	−0.0495	4.61	70.7	−0.000118
	(.0405)	(3.66)	(43.2)	(0.000317)
County FE	YES	YES	YES	YES
Farm FE	NO	NO	NO	NO
Year FE	YES	YES	YES	YES
Observations	3631	3681	3504	3694
Number of counties	1244	1244	1238	1244

a 287(g) authorization manually set to five years earlier than actual authorization; outcome variables limited to those available at the county level in 1997 b Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses c Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. d *** p<0.01, ** p<0.05, * p<0.1 e Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C14 shows another falsification test: the impact of reduced-form jail capacity in the pre-treatment period on our outcomes of interest in those years. Here, the coefficients are very small, insignificant for acres operated and number of farms in a county, and only weakly significant for labor expenses. We are also able to explore whether there are certain 287(g) counties that are driving our results by randomly dropping 10% of the treated sample and 10% of the control counties and re-running our main specification. Our findings from Table 3 are robust to bootstrapped estimates from this procedure replicated 1000 times, and there is no indication that a particular county is driving our results (see appendix Table C16).

5.3. Supplemental analyses

Together, the farm- and county-level results in Section 4.1 show evidence of a negative labor supply shock with only limited endogenous technical change. As such, farms may be making longer-term adjustments that affect their profitability and operational structure. Our ability to examine truly long-term impacts is limited by data availability; however, we look at outcomes that represent production adjustments and the financial standing of farms in Appendix B. For the former, we examine the impact of 287(g) on broad measures of acreage allocation between vegetable, fruit, and field/row crops (mechanized acres); for the latter, we use indicators of farm- and county-level agricultural sector profitability. While we find no impact of 287(g) on broad crop categories, we do find evidence of increasing vegetable production in border counties, which suggests potential labor supply spillovers. Likewise, we find no evidence of a relationship between 287(g) and farm-level indicators of profitability, but some evidence of a negative relationship at the county-level. In particular, 287(g) counties have fewer farm operations. Jointly, this provides suggestive evidence that larger, more productive farms were better able to adapt to the a higher-labor cost environment.

6. Conclusion

This study provides a unique and robust analysis of how enhanced local immigration enforcement affected US firms in a sector with a high level of undocumented immigrant labor. We use an unbalanced panel of farm survey data and a balanced panel of county-level agricultural Census data to test the degree to which endogenous technical change occurs in response to local labor supply shocks in US agriculture. We consider the impact of county-level participation in the Delegation of Immigration Authority 287(g) program, which through authorization alone had a strong deterrent effect on undocumented workers (Kostandini et al., 2014). We developed a novel instrumental variable to address the endogeneity of participation in the 287(g) program: pre-program county jail occupancy, which is strongly correlated with 287(g) participation and arguably exogenous from farm sector trends.

We find that county enrollment in 287(g) leads to increased fuel expenditure at the farm and county level across our primary specifications, although more urbanized counties may be driving this result. In all specifications, we find a county-level increase in labor expenditure and a decline in the number of workers and land in farms. Jointly, these results imply (1) consistent with other studies, 287(g) caused a negative labor supply shock; (2) farms that remained in operation likely used machinery more intensively; and (3) limited substitutability of immigrant farm workers, as indicated by the decline in land in farms and number of farm workers in 287(g) counties. The effect on labor expenditure likely reflects the confounding influences of decreasing labor supply, higher wages, or increased use of technical or managerial labor; this further suggests that limited substitutability of immigrant labor leads to non-marginal changes in labor and capital use. Comparison of the impacts measured with farm-level versus county-level data reflect different survey populations and methodologies. In our supplemental analysis, we find a decline in the number of farms in program counties; this, along with the farm-level results, indicate that larger farms were better able to absorb this shock. Overall, we see limited evidence for endogenous technical change in response to enhanced local immigration enforcement. Results are largely consistent under alternative measures of jail occupancy, 287(g) enforcement, and control group composition. Finally, the possibility of anticipatory behavior means that our post-enrollment estimates may underestimate the true impact of the program.

Our data and empirical strategy preclude estimation of labor supply or demand elasticities. Available data unfortunately provide insufficient information on agricultural wages, worker types, or meaningful output quantities. We also do not observe specific production technologies and measuring mechanization is noisy. As such, we use production expenditures, acres operated, and number of workers, which are the best available direct measures of labor supply shock responses. Further, our study cannot predict what would happen in the event of a nation-

[26] Another concern is that it is not a characteristic of the counties in question that are driving results but instead that the results are the product of widespread trends in agriculture, or unobserved characteristics of the time in which these programs were enacted. To address, though not fully refute, this claim, we performed a similar placebo test in which the timing of program authorization was kept constant and we randomly assigned program counties across the contiguous United States to 287(g) treatment status. We randomly selected 53 counties, without replacement, and randomly assigned them the start date of an actual 287(g) county. The validity of this exercise is questionable for two reasons; first, because the 2SLS estimation using these counties does not have a strong first stage. Second, this method falsely assumes an equal likelihood of 287(g) enrollment across all counties. Nonetheless, it was a useful exercise to highlight a lack of general, country-wide economic trends behind our results. This process was replicated 500 times, with nothing to indicate more than spurious results.

wide change in immigration policy–external validity is limited by the geographic scope of 287(g) participation. A final caveat is that we estimate the local average treatment effect: our estimates reflect counties whose participation in 287(g) was driven by lower jail occupancy, which in turn resulted from expanding jail capacity, reflecting an 'active' local law enforcement. As our broad research interest is local immigration enforcement as opposed to any particular program, we believe that our results provide important information about how a manual-labor-dependent sector responds to a local labor supply shock. Generally, future research would benefit from data on additional outcomes, including wages and production quantities, and analyses of different policies in the US and other countries. In addition, our instrumental variable approach may be useful in future studies on law enforcement and immigration policy; further, the first stage may be relevant to scholars of criminal justice and local government.

Immigration policy will likely continue to be politically contentious across many countries, i.e. Coddington (2019); Grande et al. (2019); McKeever (2020); Newton (2018). In the US, it is uncertain when a political agreement will emerge to provide more certainty for sectors with high levels of undocumented labor. Our results imply that the agricultural sector did, to some degree, adapt to increased local immigration enforcement, but they also provide strong evidence that neither technology nor native workers are complete substitutes for immigrant farm workers. This implies that locations with relatively stringent immigration enforcement may be less competitive in the production of labor-intensive crops. Broader, national policy changes such as the bracero exclusion considered by Clemens et al. (2018) would likely drive large changes to US agriculture. Current investments in robotics (Seabrook, 2019) suggest that the tight domestic labor market and current immigration policies may be pushing innovation for the sector as a whole. While mechanization and robotics increasingly characterize the global economy, labor shortages specific to particular industries or specializations are likely The impact of these shortages on firms is a crucial part of the political and economic debate on imto endure through economic and political changes.migration policy.

Appendix A. Exclusion restriction analysis

In this supplementary appendix, we have selected the most plausible possible confounders and provide evidence that an invalidating relationship does not exist between these measures and our instrument, outcomes, and treatment variables. The validity of our instrumental variable rests on whether or not the exclusion restriction is satisfied, or that jail occupancy affects farm outcomes *only* through its impact on immigration enforcement. Part of defending the exclusion restriction involves eliminating the possibility that there are confounding variables that impact both the instrument itself and the outcome variables of interest. While it is impossible to prove that we have tested every potential variable that could be related to our IV (jail occupancy) and in turn affects both our treatment (287(g) program participation) and our farm or agricultural sector outcomes of interest, we have conducted an exhaustive analysis of factors that could hypothetically be related to farm outcomes and jail occupancy.

Recent papers using instrumental variables have demonstrated that the presence of a covariate that is determined by the instrumental variable *and* affects the endogenous treatment as well as the outcome in question constitutes a violation of the exclusion restriction. For example, Faber and Gaubert (2019) estimate the relationship between tourism activity and local socio-economic outcomes in Mexico, using pre-existing geologic and geographic features as an instrument for tourism activity. They find, however, that the presence of those features also increased the likelihood of an area receiving public investment from a national program; this investment in turn affected both tourism activity and the outcomes of interest. Mellon (2021), shows that weather, while often used as an instrumental variable due to its exogeneity, does not satisfy the exclusion restriction, because the literature has shown weather to be related to a myriad of factors that in turn are related to the outcomes considered in these studies.

Table A1
Relationship between instrumental variable (jail occupancy) and county-level covariates.

	(1) Jail occupancy IV	(2) Jail occupancy IV
Average wage, manufacturing sector	0.000969	
	(.000596)	
Average wage, service sector	–0.00162	
	(0.00167)	
Educational attainment: Less than grade school	10.5	
	(617)	
Educational attainment: Completed grade school	–242	
	(260)	
Educational attainment: Completed high school	95	
	(77.9)	
Educational attainment: Some college or more	207	
	(229)	
Percent of population below poverty line	–165	
	(223)	
Manfacturing employment (%)	204	
	(155)	
Service Employment (%)	–218*	
	(123)	
Farm dependent		–0.488
		(0.517)
Mine dependent		0.353
		(1.06)
Manufacturing dependent		–0.417
		(0.53)
Federal/state gov't dependent		–3.25***
		(1.11)
Services dependent		2.82
		(2.81)
Housing stress		2.89
		(1.97)
Low education		-1.16
		(1.27)
Low employment		–0.34
		(1.29)
Persistent poverty		-1.32
		(2.32)
Population loss		0.318
		(0.715)
Non-metro recretation destination		0.019
		(1.11)
Retirement destination		-2.24*
		(1.3)
Persistent child poverty		2.99
		(2.71)
Constant	7.4*	0.948*
	(4.3)	(0.517)
Observations	10,944	10,944
R-squared	0.0131	0.0196
Number of counties	2763	2763

a Standard deviations in parenthesis b ***,**,*: Significantly different from 287(g) counties at 1%, 5%, and 10%, respectively

To start, we show that there is no evidence for anything other than a spurious relationship between our instrument and two sets of county-level covariates. Table A.1 presents results from a specification that has our instrument as the dependent variable; the covariates in column (1) are indicator variables from the 2004 County Typology data from the USDA Economic Research Service (ERS), which characterize a county's sectoral economic dependence and other policy-relevant features at one point in time. As these indicators do not change during the period we study, we provide additional evidence that our instrument is not related to other county-level economic factors. Column (2) shows the results from a specification with our instrument as the dependent variable and time-variant measures of a county's socio-economic status on the right-hand side.

Table A2
County typology comparison between 287(g) and non-287(g) counties.

	No. 287(g) mean	287(g) mean	287(g) border mean
County is:			
farm dependent	0.142*** (0.349)	0 (0)	0.050*** (0.219)
mine dependent	0.0412*** (0.199)	0 (0)	0.010***(0.010)
manufacturing dependent	0.289*** (0.453)	0.211 (0.409)	0.352*** (0.478)
federal/state government dependent	0.121 (0.327)	0.0916 (0.289)	0.176*** (0.381)
services dependent	0.105*** (0.307)	0.426 (0.496)	0.191*** (0.393)
nonspecialized dependent	0.302 (0.459)	0.271 (0.445)	0.211* (0.408)
a non-metro recreation destination	0.106*** (0.308)	0.0558 (0.230)	0.0905** (0.287)
a retirement destination	0.139*** (0.346)	0.287 (0.453)	0.281 (0.450)
County has:			
housing stress	0.168*** (0.374)	0.398 (0.491)	0.186*** (0.389)
low education	0.199*** (0.400)	0.0876 (0.283)	0.166***(0.372)
low employment	0.148*** (0.355)	0.0199 (0.140)	0.0704*** (0.256)
persistent poverty	0.124*** (0.330)	0 (0)	0.0452*** (0.208)
population loss	0.193*** (0.395)	0 (0)	0.0201*** (0.140)
persistent child poverty	0.236***(0.425)	0.0319 (0.176)	0.161*** (0.367)
n	3093	50	199

a Standard deviations in parenthesis b ***,**,*: Significantly different from 287(g) counties at 1%, 5%, and 10%, respectively

As Table A.1 shows, there is no strong or systematic relationship between either set of these variables and our instrument. Because the instrument is scaled by jail capacity, it takes into account a county's projected needs in terms of incarcerated populations. Scaling by jail capacity takes into account that there are counties with larger populations and/or more crime; these counties almost always have higher jail capacities. This explains at least to some extent why there is no significant relationship between our instrument and the measures in columns (1) and (2); many of these factors are likely strongly related to the *absolute* size of a county's jailed population. With that being said, the relationship between many of these factors and the performance of a county's agricultural sector is quite low. Agricultural operators make up such a small share of both county population and GDP that even if there were evidence for a systematic relationship in Table A.1, it would be difficult to then connect these outside factors to our agricultural outcomes of interest.

A1. County economic specialization

Although we find no evidence that our instrument is related to any of these county typology codes, we can use them to ascertain whether there are any systematic differences between program counties and non-program counties that would help identify other potential confounding factors or omitted variables. While these variables are implicitly controlled for through the farm or county-level fixed effects in all our specifications, they can indicate potential confounders that need to be further analyzed. In order to disrupt our identification strategy, these factors would have to explain both 287(g) program participation and the evidence for a contracting agricultural sector in these counties. We therefore compare 287(g) counties to the rest of the country in 2004, before 287(g) adoption in any county. This comparison is summarized in Table A.2, which shows that while many differences exist between 287(g) counties and counties without the program, these differences generally do not fit the narrative that would compromise the exogeneity of the instrument by providing evidence of an omitted outside factor that explains the relationship we have identified.

Most importantly, these data indicate that none of the 287(g) counties are agriculture-dependent.[27] Although weak to begin with, the connection between economic trends at large and trends in the farm economy are likely to be strongest in counties considered agriculture dependent. As no counties that participated in 287(g) could meet even this low threshold for dependence on agriculture, it is even less likely that general economic conditions affecting jail occupancy or renovations in a county would also be impacting agriculture or vice versa.

Program counties are also not experiencing population or economic decline, which could lead to emptier jails and farm sector contractions. No 287(g) county experienced population loss, defined in the 2004 (2015) County Typology data as a significant decline in the county population between the 1990 and 2000 (2000 and 2010) US Census. Further, only 2% of 287(g) counties are classified as low-employment counties, where a low-employment county is one where less than 65% of the working age population in the county is employed. Hopkins (2010) found that counties with increasing unemployment were more likely to enact anti-immigrant ordinances; what we see here is actually the opposite, in that most 287(g) counties did not experience widespread unemployment. These classifications, therefore, provide some evidence that these are not a subset of counties doing significantly worse economically than other counties, and neither are they counties where poor outcomes in the agricultural sector are likely to affect outcomes for the county's non-farm economy.

A related potential confounding factor is that increases in housing values, or in other non-agricultural property values, could reflect increased development pressure, negatively affecting farm asset values and operational costs as well as being related to broader economic trends that influence law enforcement. Fig. A.1 shows the trend across the study period for properties with and without a mortgage. We see property values in the treated and border groups actually decline significantly during this time, indicating that development pressure is unlikely to be driving the observed relationship.

These data also show that there is not unique development happening in program counties relative to their neighboring ones: there is no statistical difference in the trend in any year. Tests of the difference in the means of median housing values, for properties with and without a mortgage, between 287(g) counties and border counties also reveal no significant differences.

[27] According to the ERS, a county is agriculture-dependent if it meets one of two criteria: either 1) farm earnings account for an annual average of 15 percent or more of total county earnings during 1998–2000 or 2) farm occupations account for 15 percent or more of all occupations of employed county residents in 2000. See https://www.ers.usda.gov/data-products/county-typology-codes.aspx for more information on these data.

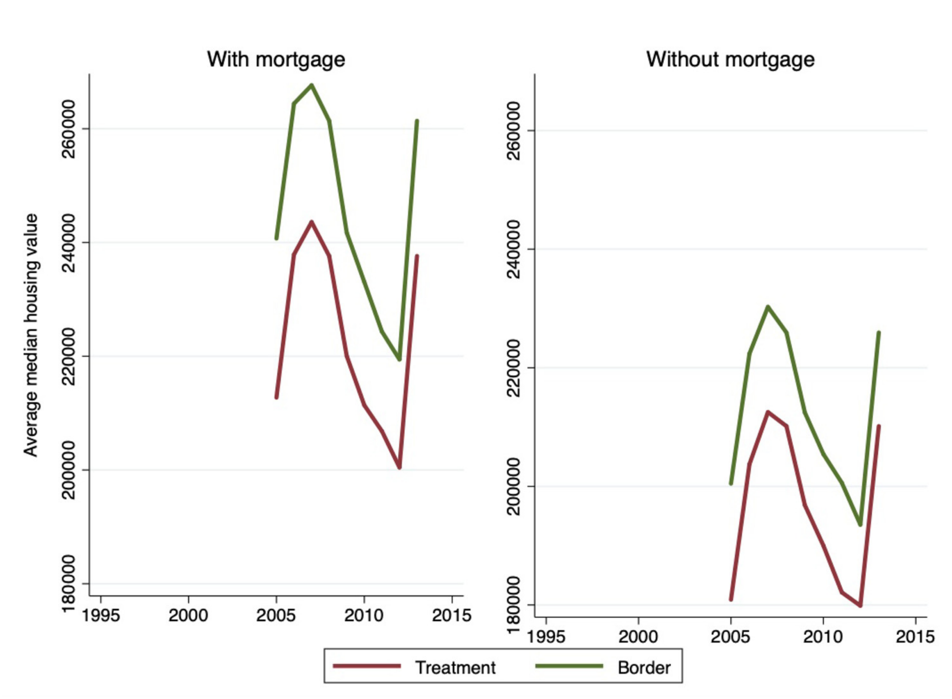

Fig. A1. Housing values followed a similar trend in program and border counties.

A2. Weather

As the thorough review in Mellon (2021) readily demonstrates, the impacts of weather are myriad and far-reaching. Weather patterns, including trends of increasing temperatures, may affect jail occupancy as well as farm outcomes. The lumpiness of jail capacity data makes analyzing the relationship between our instrument and measures of weather statistically challenging: while we have access to monthly or even daily weather for each county, jail capacity is measured only once a year. This complication actually strengthens the exclusion restriction: because the counties that enacted 287(g) are spread across the country, the growing seasons differ in each county. In order for weather to have an impact on both our instrument and agricultural outcomes, it would have to affect different counties with different weather patterns and agricultural systems in the same way, while affecting border counties differently than their neighboring program counties.

Recent studies suggest that there is a strong correlation, if not a causal relationship, between an area's temperature and crime levels. Ranson (2014), for example, finds a strong positive effect of temperature on criminal behavior over the last 30 years, using US county-level data. However, yields typically decline with temperature (Rosenzweig et al., 2001). Likewise, greater precipitation, which dampens levels of crime according to some studies, increase yields for agricultural commodities. As such, even if weather had some impact in each county on jail occupancy, the weather events driving an area's jailed population differ from those driving its agricultural outcomes. This potential effect would bias downwards any impact of 287(g) on agriculture in the context of our empirical strategy: increasing temperatures, especially in the southern part of the US where 287(g) programs are concentrated, would inhibit rather than enhance agricultural production. Thus, there is limited theoretical evidence that weather is an omitted, confounding factor that violates our exclusion restriction.

A3. Farm workers and crime

Another potential violation of our exclusion restriction would exist if there was some direct impact of jail capacity on agricultural outcomes, such as if jail capacity was changing because farmworkers wer being incarcerated. There is little evidence to suggest that farm workers, whether native or immigrant, commit crimes with any more frequency than other members of the population. In fact, agriculturally intensive counties have significantly lower rates of virtually all crimes for which statistics are reported, although this may be due to their being rural rather than farm sector intensity. Nonetheless, comparison of the means across these two groups, for a selection of reported crimes from the Uniform Crime Reporting Program Data (US Department of Justice, 2014), can be found in Table A.3.

Extensive research on the relationship between immigration and crime has been conducted (Bell et al., 2013), with most empirical studies finding no relationship between increased immigration to an area and the crime rate. Because many of these papers who examine this relationship in the US use an IV approach where the population of immigrants from a particular country is the instrument, and these data are generally only available for urban areas, their focus is on the impact in cities.

Table A3
Summary statistics: Crime rate comparison between agriculture-dependent and non-dependent counties.

	Non-farm dependent mean (sd)	Farm dependent mean (sd)
Murder	4.566*** (24.07)	0.418(2.322)
Rape	8.995*** (30.07)	0.902(3.994)
Robbery	35.83*** (195.4)	1.327 (6.975)
Assault	148.0*** (760.1)	13.24 (94.54)
Burglary	91.31*** (346.2)	9.175 (35.00)
Larceny	377.9*** (1140)	20.36 (66.82)
Motor vehicle theft	46.91*** (301.5)	3.191 (16.78)
Arson	5.296*** (15.99)	0.466 (1.497)
Weapons charges	52.53*** (232.5)	4.041 (23.74)
Drug violations	507.5*** (2206)	45.05 (257.0)
Liquor-related violates	217.2*** (678.2)	25.98 (61.37)
Disorderly conduct	229.9*** (841.8)	13.97 (29.67)
Vagrancy	9.560*** (93.39)	0.268 (1.315)
n	2704	340

a Standard deviations in parenthesis b ***,**,*: Significantly different from farm dependent counties at 1%, 5%, and 10%, respectively

However, as Bell et al. (2013) point out, immigrant workers may experience an extra disutility from crime compared to native workers, as they face the additional penalty of deportation. This is especially true for undocumented workers, who make up a large share of the agricultural workforce.

Recent studies provide no evidence that immigrant farm workers drive crime-related trends. Hopkins (2010) finds that counties with higher crime rates are actually less likely to consider anti-immigrant ordinances. Likewise, the Secure Communities program, which allowed for the Federal government to check the immigration status of anyone locally arrested, had no impact on local crime levels (Miles and Cox, 2014). Bianchi et al. (2012) found, using data from Italy from 1990 to 2003, that a higher immigration population did not cause an increase in the crime rate. Robberies did increase, but were a small share of all criminal activity. Hines and Peri (2019) find that increase in deportations under the Secure Communities program did not affect violent or property offenses. Thus, based on available evidence, it is unlikely that an increasingly criminal immigrant farm labor force was driving both agricultural outcomes and jail occupancy or renovations during our study.

Appendix B. Analysis of additional measures of 287(g) impact

B1. Acreage allocations

Together, the farm- and county-level results in Section 5.1 show evidence of a negative labor supply shock with limited endogenous technical change. As such, farms may be making longer-term adjustments

Table B1
Summary statistics: Acres harvested, pre-287(g).

	All			287(g)			Non-287(g)		
	mean	sd	n	mean	sd	n	mean	sd	n
Vegetable acres [a]	28.8	334.8	11,730	24.5	217.3	439	29.0	338.6	11,291
Vegetable acres [b]	1,915.8	12,128.8	1,123	2,776.9	7,152.9	51	1,874.9	12,316.1	1,072
Fruit acres [a]	78.8	3,007.6	11,730	19.3	225.6	439	81.1	3,065.1	11,291
Mechanized acres [a]	545.6	1,091.5	11,730	286.8***	847.2	439	555.6	1,098.7	11,291

c ***,**,* Significantly different from non-287(g) counties at 1%, 5%, and 10% respectively
d Source: US Census of Agriculture (NASS QuickStats); USDA Agricultural and Resource Management Survey (ARMS)
[a] Outcome from ARMS
[b] Outcome from Census

Table B2
Impact of 287(g) authorization on farm acreage decisions: 2SLS .

	(1) Vegetable acres (ARMS)	(2) Vegetable acres (Census)	(3) Fruit acres (ARMS)	(4) Mechanized acres (ARMS)
287(g) authorization	34.58	-0.031	1735	71.91
	(41.91)	(0.115)	(1,975)	(95.03)
287(g) border	16.57**	0.008	116.9	-8.984
	(7.352)	(0.005)	(138.7)	(11.45)
County FE	NO	YES	NO	NO
Farm FE	YES	NO	YES	YES
Year FE	YES	YES	YES	YES
Observations	21,403	4316	21,403	21,403
Number of counties	-	1229	-	-
Number of farms	11,501	-	11,501	11,501

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** $p<0.01$, ** $p<0.05$, * $p<0.1$
d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table B3
Summary statistics: Financial indicators, pre-287(g).

	All			287(g)			Non-287(g)		
	mean	sd	n	mean	sd	n	mean	sd	n
Machinery value[a]	$254.3	$527.8	11,730	$201.1**	$363.7	439	$256.4	$533.0	11,291
Machinery val/op.[b]	$57.8	$41.7	1,326	$47.9**	$27.0	53	$58.2	$42.1	1,273
Net income[a]	$210.7	$1,200.4	11,730	$298.6	$1,023.5	439	$207.3	$1,206.6	11,291
Net income[b]	$13,969.7	$44,599.8	1,314	$29,264.4**	$46,922.4	53	$13,326.9	$ 44,403.7	1,261
Real estate assets[a]	$2,126.9	$10,500.0	11,730	$2,683.8	$7,455.9	439	$2,105.3	$10,600.0	11,291
Real estate assets[b]	$ 371,315.9	$ 518,338.0	1,325	$546,440.7**	$570,682.9	53	$364,019.0	$514,996.4	1,272
Debt (ARMS) [a]	$361.5	$1,463.7	11,730	$329.7	$1,099.3	439	$362.7	$1,476.1	11,291
# of farms[b]	494.4	474.2	1,328	664.5**	560.7	53	487.3	469.1	1,275

c Dollar values are in thousands of dollars.
d***,**,* Significantly different from non-287(g) counties at 1%, 5%, and 10% respectively
e Source: US Census of Agriculture (NASS QuickStats); USDA Agricultural and Resource Management Survey (ARMS)
[a] Outcome from ARMS
[b] Outcome from Census

Table B4
Impact of 287(g) authorization on financial indicators: 2SLS.

	(1) Machinery value (ARMS)	(2) Machinery value per operation (Census)	(3) Net income (ARMS)	(4) Net income per county (Census)	(5) Real estate asset values (ARMS)	(6) Real estate asset values per county (Census)	(7) Debt (ARMS)	(8) Number of farms per county (Census)
287(g) auth.	203,898	18.4	-272,655	-1,920**	-4,395,000	31,936***	847,386	-0.015***
	(278,104)	(13.2)	(460,024)	(940)	(2,636,000)	(10,543)	(929,858)	(0.005)
287(g) border	-3,287	2.64	38,458	-104	-346,501	1,692***	-23,829	-0.0006
	(12,087)	(1.72)	(76,102)	(70.5)	(317,702)	(543)	(31,119)	(0.0004)
County FE	NO	YES	NO	YES	NO	YES	NO	YES
Farm FE	YES	NO	YES	NO	YES	NO	YES	NO
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Observations	21,403	4,942	21,403	3,542	21,403	4,940	21,403	4,944
# of counties	-	1,239	-	1,238	-	1,239	-	1,239
# of farms	11,501	-	11,501	-	11,501	-	11,501	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

that affect their profitability and operational structure. Our data also allow us to consider 287(g) impacts on substitution between very coarse crop categories and impacts on farm viability in the short term. We estimate the impact of the 287(g) program on acreage allocations: vegetable acres, fruit acres, and mechanized acres.[1] This effectively disaggregates the "acres operated" outcome into possible component parts.

Although not a direct test of the theory of endogenous technical change due to insufficiently granular information on crop type, we examine whether there is evidence of operations shifting acreage between the broad production specializations available in our data, which may to some degree reflect increased use of automatic technology. While many fruits and vegetables can be mechanically harvested, others cannot; other crops, like field or row crops, are almost entirely mechanically harvested. To evaluate the degree to which farmers are able to switch to less labor-intensive crops, we aggregate acres for all crops individually reported in ARMS that are typically mechanically harvested.[2] Vegetable and fruit production is generally more labor-intensive, so a decline in the number of vegetable or fruit acres harvested may reflect less labor use. Similarly, an increase in 'mechanized crops' would suggest substitution between different types of production. Our data do not allow us to capture a shift from manual to automatic technologies within the fruit and vegetable category, which is where we would expect that most farms adjust in response to a labor supply shock.

We find no statistically significant change in acres of fruit, vegetables, or mechanized crops harvested, shown in Table B2. This may reflect our inability to distinguish between different kinds of fruits and vegetables, let alone between production of a specific crop for processing or for the fresh market. Such distinctions capture the degree to which fruit and vegetable production is mechanized. Our null result for 'mechanized acres' may reflect the high cost associated with changing production systems, or the agronomic suitability of more fixed assets (namely, land) playing a primary role relative to labor costs or shortages. Further, we also observe that farms in counties adjacent to a 287(g) county experienced a statistically significant increase of 17 vegetable acres harvested, which is about half of the average non-287(g) county production. Hypothetically, 287(g) authorization could have deterred immigrant labor in adjacent counties or encouraged workers to move to these bordering counties. This result provides some suggestive evidence that there may have been positive labor supply spillovers associated with 287(g). In other words, undocumented workers may have moved to nearby counties, where production practices are similar, but do not have the punitive 287(g) authorization programs in place. This may have allowed farms in bordering counties to expand production levels without increasing expenditure.

B2. Longer term impacts

The endogenous technological change framework suggests a decline in production in terms of output and land use if labor and technology are not substitutable, or if substitution is incomplete. This could lead to a decline in firm profitability and, at its extreme, impact firm survival. We augment our ability to predict the long-term effects of labor supply shocks on the farm economy using a set of variables that capture the financial health of the farm business. Changes in these variables point to larger structural changes in the local farm sector. In particular, number of farms captures the most extreme outcome, farm exit, that can occur as a result of a challenging farm business environment. Both net farm income and asset value assess whether the changing production environment is reflected in farms' financial status and asset holdings. Farm asset values reflect market expectations for long-term profitability of the farm business, with labor shortages potentially being capitalized into asset values. Changing machinery values could reflect a shift to automatic technologies, as well as changes to the quantity of machinery and equipment owned by farm operations.

Although no truly long-term impacts can be considered given the time frame, we have data from up to five years following 287(g) implementation. Results of our analysis of the relationship between 287(g) and various indicators of farm financial status are reported in Table B4. For the farm-level data, we do not find strong evidence that the relationship between 287(g) and farm machinery value, net income, real estate asset value or debt is statistically different than zero. For these ARMS farms, sampled before and after 287(g) implementation, this provides no support for broader financial impacts for operations that existed before and after 287(g) implementation. We do see suggestive evidence of a weakening farm economy in these counties: program adoption is associated with fewer farms in a county as well as lower net farm income.

[1] All of these are calculated from the ARMS data and thus are farm-level; of these, only vegetable acres harvested is reported in the Census. Regardless of source, they are only reported in terms of acres harvested, rather than operated or planted.

[2] These crops are barley, canola, corn, cotton, hay, oats, peanuts, potatoes, rice, sorghum, soybeans, sugar, tobacco, and wheat.

Appendix C. Additional tables and figures

C1. Figures

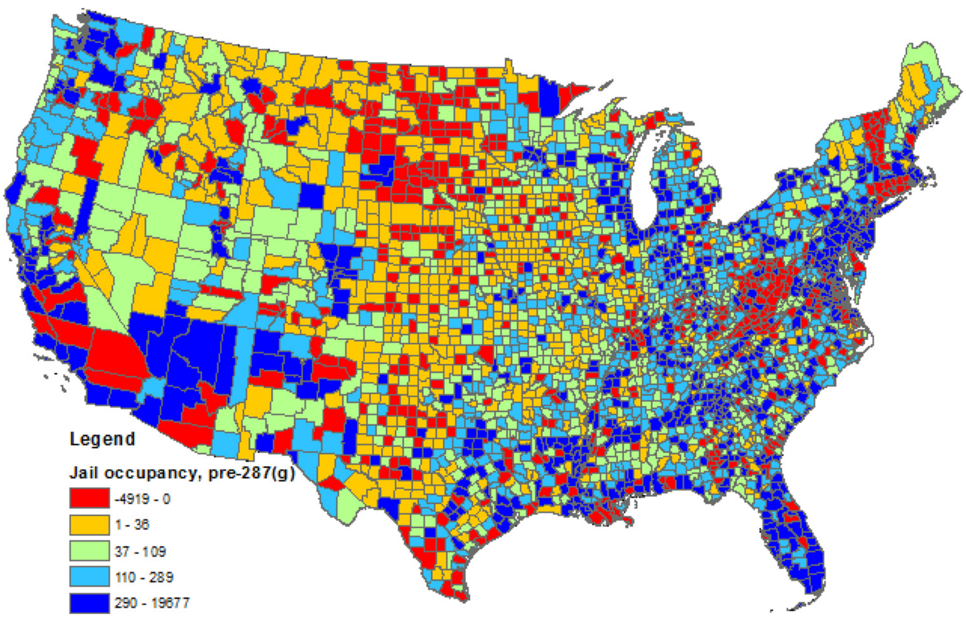

Fig. C1. US jail occupancy, 2005–2006.

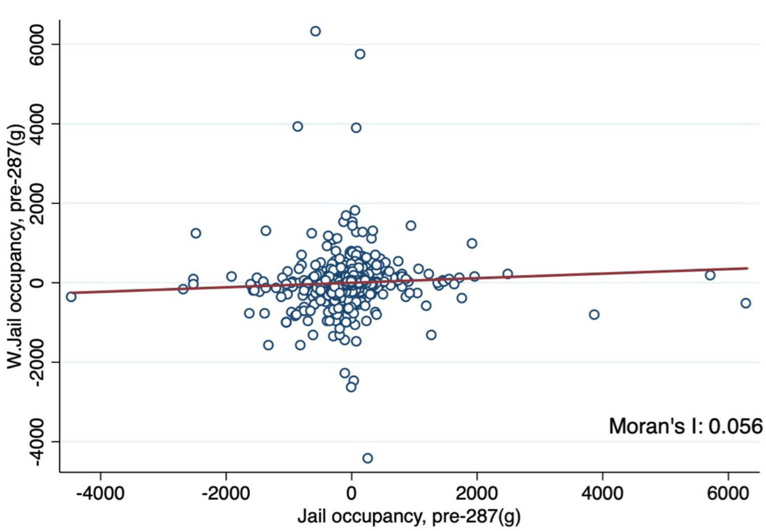

Fig. C2. Moran's I rejects spatial correlation for jail occupancy.

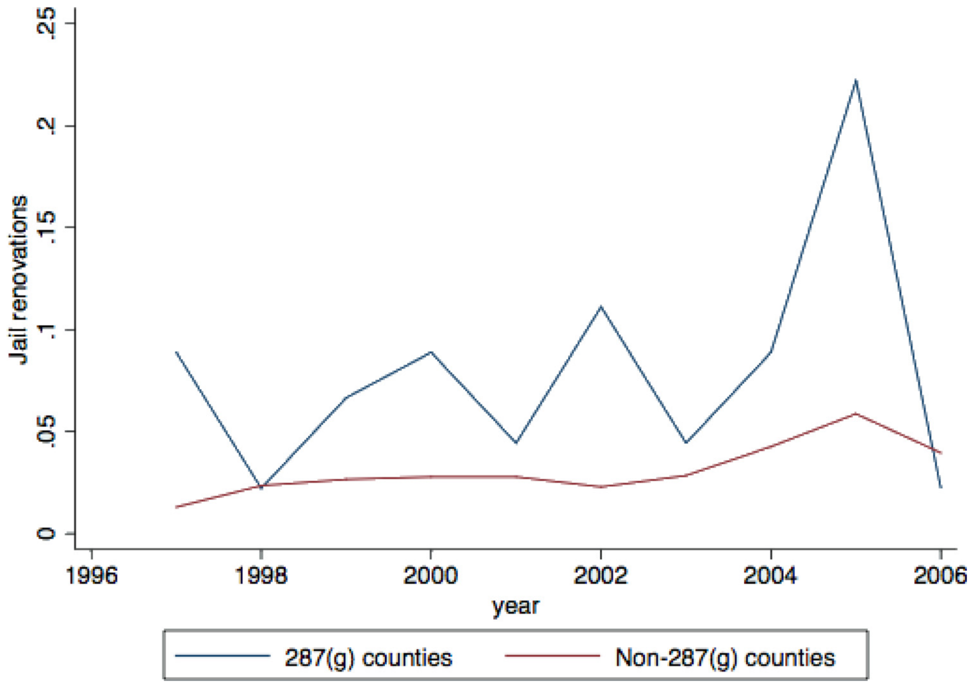

Fig. C3. 287(g) counties were investing more in local jails since 1996.

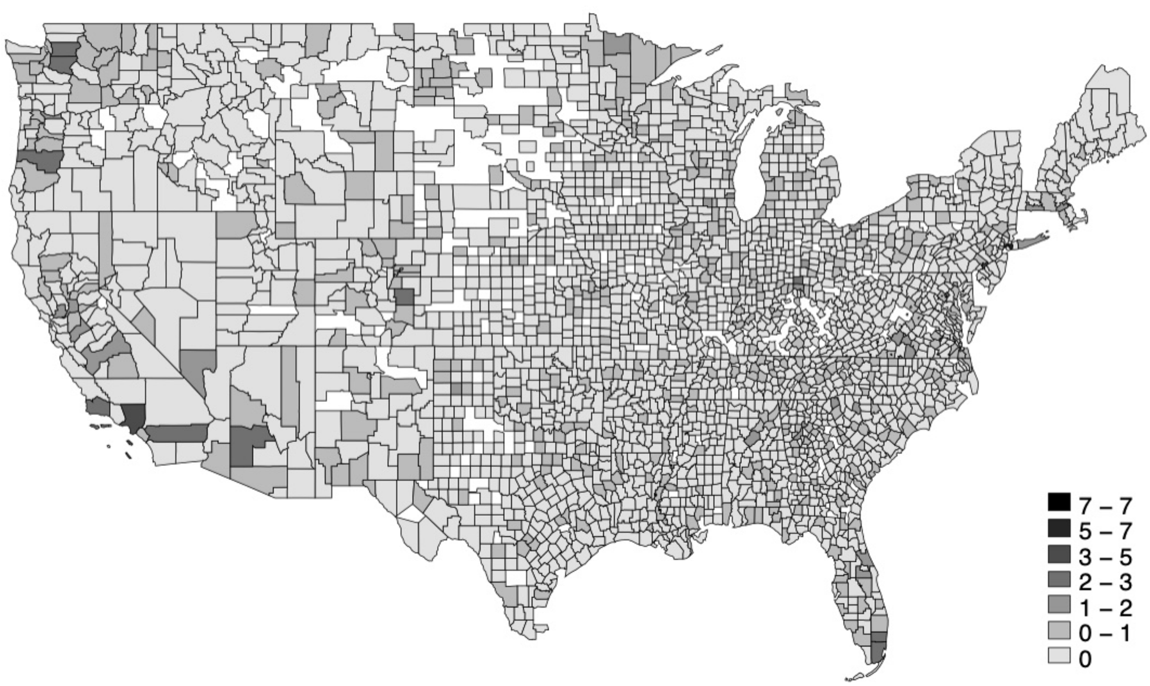

Fig. C4. US jail renovations, 1997–2006.

C2. Summary statistics

Table C1
Summary statistics: 287(g) participation and jail occupancy.

	287(g)			Non-287(g)		
	mean	sd	n	mean	sd	n
287(g) authorization: all years	33.5%	47.3%	212	0%	0%	5100
287(g) authorization: 1997	0%	0%	53	0%	0%	1275
287(g) authorization: 2002	0%	0%	53	0%	0%	1275
287(g) authorization: 2007	39.6%	49.4%	53	0%	0%	1275
287(g) authorization: 2012	94.3%	23.3%	53	0%	0%	1275
287(g) enforcement[a]	57.5%	49.7%	106	-	-	0
Year 287(g) program authorized[a]	2007.6	0.9	106	-	-	0
Aliens identified[a]	457.6	908.2	106	-	-	0
Aliens departed[a d]	258.2	596.5	106	-	-	0
Jail occupancy pre 2007	-121.6	1006.9	212	-22.2	306.9	5100

b Sample is restricted to counties in states that have a jurisdiction with a 287(g) program
c Source: ICE FOIA Proactive Disclosures; Bureau of Justice Statistics (BJS) National Jail Census
[a] Conditional on having 287(g) program authorization
[d] d "Aliens departed" is the official language used by ICE

Table C2
Summary statistics: difference in pre-287(g) trends between 287(g) and non-287(g) counties.

	Non 287(g)	287(g)	287(g) border
		Mean difference	
Census of Agriculture data			
Fuel expenditure	0.98	0.97	0.84
	(0.94)	(0.71)	(0.93)
Number of farms	0.09	-0.01	0.08
	(0.63)	(0.24)	(0.6)
Acres in farms	-0.05**	-0.22	-0.06
	(0.44)	(0.6)	(0.62)
Asset value per operation	0.21*	-0.01	0.11
	(0.75)	(0.51)	(0.51)
Machinery Value per operation	0.25	0.23	0.15
	(1.09)	(0.67)	(0.64)
Other county data			
Population	-0.02*	-0.04	-0.04
	(0.07)	(0.06)	(0.07)
Average manufacturing wage	-1.07	-1.05	-1.34
	(0.82)	(0.93)	(1.83)
Average service wage	-0.04	-0.19	-0.27
	(0.70)	(0.83)	(1.17)
% No school	-0.42	-0.44	-0.57
	(0.46)	(0.68)	(0.70)
% Grade school	-0.11	-0.14	-0.21
	(0.36)	(0.43)	(0.52)
% High school	-0.37	-0.32	-0.45
	(0.49)	(0.54)	(0.66)
% One year of college or more	0.03	0.05	-0.08
	(0.45)	(0.49)	(0.81)
% Below poverty line	-0.14	-0.26	-0.29
	(0.48)	(0.72)	(0.74)
% Employed in manufacturing	-0.67*	-0.86	-0.82
	(0.74)	(0.47)	(0.79)
Number of counties	995	44	179

a Standard deviations in parenthesis
b ***,**,*: Significantly different from 287(g) counties at 1%, 5%, and 10%, respectively
c Sample is restricted to counties in states that have a jurisdiction with a 287(g) program
d Sources: USDA Census of Agriculture, US Census and American Community Survey
e Note: Difference in percent change between 1997-2002 and 2007 for Census of Agriculture data; 1998-2002 and 2002-2006 for population and 1990-2000 and 2000-2007 for all other data.
f Percent of county population with at least one year of education at level specified and at most a diploma at level specified

Table C3
Summary statistics: Jail construction and renovation.

	No. 287(g)			287(g)			287(g) border		
	mean	sd	n	mean	sd	n	mean	sd	n
Avg. year jail construction	1980.2	23.3	2,812	1987.8***	10.4	140	1984.7***	15.8	312
Avg. year jail renovation	1997.5	8.7	1,132	1999.0	5.9	94	1997.7	7.3	162
# of jails constructed, 1997–2006	0.30	0.63	2,812	0.53***	0.72	140	0.34	0.51	312
# of jails renovated, 1997–2006	0.28	0.50	1,132	0.81***	1.11	94	0.43***	0.71	162
# of jails constructed, 2001–2006	0.14	0.45	2,812	0.25**	0.49	140	0.14	0.34	312
# of jails renovated, 2001–2006	0.19	0.42	1,132	0.55***	0.87	94	0.25	0.53	162
# of jails constructed, 2006	0.00	0.04	2,812	0***	0.00	140	0.01	0.10	312
# of jails renovated, 2006	0.04	0.21	1,132	0.03***	0.16	94	0.05	0.25	162

a ***,**,*: Significantly different from "neither" counties at 1%, 5%, and 10%, respectively b Sample is restricted to counties in states that have a jurisdiction with a 287(g) program c Source: BJS National Jail Census, 2006

Table C4
Summary statistics: Production outcomes by availability of jail renovation data.

	No jail renovation data			Jail renovation data		
	mean	sd	n	mean	sd	n
Acres operated	103,917.28*	106,595.59	925	115,137.94	184,768.99	1523
Number of farms	446.65***	342.59	945	487.65	419.73	1560
Number of workers	786.58***	1,110.73	943	1,102.78	3,435.36	1550
Net income	$11,969.0	$ 17,338.4	934	$ 13,445.7	$ 35,941.2	1548
Asset value	$337,551.9***	$ 274,243.0	943	$ 401,138.2	$ 459,549.6	1558

a ***,**,*Mean significantly different from mean of counties with jail renovation data at 1%, 5%, and 10%, respectively b Sample is restricted to counties in states that have a jurisdiction with a 287(g) program c Dollar values are in thousands of dollars. d Source: Bureau of Justice Statistics (BJS) National Jail Census, 2006

C3. Results

This specification uses all counties in the contiguous United States in the control, rather than limiting the control group to untreated, non-adjacent counties in states with a 287(g) jurisdiction, as in our main specification. Because program participation was based on self-selection, every county in the United States had the opportunity to participate: nonetheless, program participation is concentrated in counties in the southern part of the United States and does not uniformly represent all agricultural systems. Our preferred control group is limited, therefore, to take into account that program participation may not have been as viable, economically or politically, in different parts of the country.

C4. Alternate standard errors specifications

We report standard errors that are robust to correlation at the strata level for the ARMS results (following Weber and Clay (2013)) and heteroskedasticity-robust standard errors for Census results; we acknowledge that any single approach to estimating standard errors for this study requires assumptions that may not always hold. Our results are largely consistent across a number of approaches to specifying standard errors. Below, we report our main results with standard errors robust to correlation at the ERS production region and state level (tables C17 and C18, respectively), as well as county-level clusters for our Census outcomes (Table C19).

Table C5
Reduced form estimates: impact of jail occupancy on production decisions.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
	Panel A: With control for border control						
Jail occupancy, pre-2007	0.041	0.040***	–7.48***	–2.18**	-47.66	-6.61***	0.001**
	(0.042)	(0.008)	(3.422)	(1.06)	(41.16)	(2.26)	(0.0003)
287(g) border	–54.03	–0.050*	1739	10.8	-17,259	37.1**	0.006
	(78.56)	(0.028)	(3,424)	(8.56)	(27,805)	(17.3)	(0.005)
Observations	21,403	4874	21,403	4923	21,403	4690	3681
Number of counties	-	1239	-	1239	-	1237	1237
Number of farms	11,501	-	11,501	-	11,501	-	-
	Panel B: Without control for border county						
Jail occupancy, pre-2007		0.037***		–2.39**		–7.29***	0.001**
		(0.008)		(1.02)		(2.21)	(0.0002)
Observations		4874		4923		4690	3681
Number of counties		1239		1239		1237	1237
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C6
2SLS First stage: jail occupancy and renovations.

	(1) ARMS	(2) Census	(3) ARMS	(4) Census
Jail occupancy, pre-2007	0.00009***	–0.010***		
	(0.000004)	(0.003)		
Jail renovations, pre-2007			0.044***	0.046***
			(0.007)	(0.016)
287(g) border	–0.065***	–0.038***	–0.086***	–0.037***
	(0.006)	(0.008)	(0.008)	(0.008)
County FE	NO	YES	NO	YES
Farm FE	YES	NO	YES	NO
Year FE	YES	YES	YES	YES
F-stat	43.02	12.58	43.42	9.72
σ_u	0.051	0.094	0.799	0.071
σ_e	0.142	0.109	0.157	0.101
ρ	0.362	0.428	0.354	0.328
Observations	21,839	4874	14,591	4843
Number of counties	-	1239	-	1228
Number of farms	11,753	-	7858	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C7
Impact of 287(g) authorization: 2SLS with jail renovations IV.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
	Panel A: Reduced form						
Jail renovations, 1997–2006		-2,647		1,469,341**		16,456,472	-391**
		(3,137)		(693,563)		(11,837,757)	(180)
287(g) border		-5,275*		380,142		1,228,002	29.9
		(3,069)		(602,125)		(5,287,983)	(50.2)
	Panel B: With jail renovation IV						
287(g) authorization	203.3	-1.04*	103,600**	152	-302,444	597	-.0967**
	(308.8)	(.618)	(51,432)	(106)	(216,437)	(390)	(.0477)
287(g) border	-102.2	-.0931**	8893	16.5*	-52,543	60.2**	0.00216
	(119.8)	(.0403)	(8,135)	(9.84)	(56,920)	(23.6)	(.00487)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	14,591	4843	14,591	4891	14,591	4660	3657
Number of counties	-	1228	-	1228	-	1226	1226
Number of farms	7858	-	7858	-	7858	-	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

None of these levels of clustering are entirely appropriate. Agronomic conditions, weather, and cropping patterns are not constrained or contained by state (or county) boundaries, and so clustering at the state level ignores these important relationships in the dependent variable that span geopolitical boundaries. The agricultural resource regions were developed in part to address this: the regions disregard state boundaries and look only at counties that are agronomically and agriculturally similar. They are limited in their suitability, however, as resource regions contain many states: there is likely within-state correlation of standard errors related to law enforcement practices, agricultural or legal policies, and immigrant populations. Further, at the region level, an insufficiently large number of clusters becomes a concern.

We also report Conley spatial standard errors for Census data, which are robust to correlation within 200km from the county centroid. The Conley spatial standard errors are not designed to be used with an IV, so the results are reported from manual 2SLS in Table C20. Our main

Table C8
Impact of 287(g) authorization: 2SLS with jail occupancy and renovations IVs.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
				Panel A: Reduced form			
Jail occupancy, pre-2007		0.035***		-2.09*		-6.21***	0.0007**
		(0.008)		(1.08)		(2.32)	(0.0003)
Jail renovations, 1997–2006		–0.044*		6.62		26.7*	-0.004**
		(0.023)		(4.17)		(14.5)	(0.002)
287(g) border		-0.049*		10.7		37**	0.006
		(0.028)		(8.58)		(17.3)	(0.005)
				Panel B: 2SLS, with both jail occupancy and jail renovation IVs			
287(g) authorization	289.4	–1.49**	152,368***	167*	121,573	616*	-0.092**
	(365.0)	(0.647)	(50,034)	(93.7)	(189,905)	(337)	(0.038)
287(g) border		–0.111***		17*		60.9***	0.002
		(0.040)		(9.42)		(22)	(0.005)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	14,571	4832	14,571	4879	14,571	4649	3650
Number of counties	-	1228	-	1228	-	1226	1226
Number of farms	7846	-	7846	-	7846	-	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C9
Impact of 287(g) enforcement levels: Aliens identified and departed, 2SLS.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
				Panel A			
Aliens identified	–0.0714	–0.0048***	12.94**	0.298*	82.49	0.875**	-0.000108**
	(0.0729)	(0.00179)	(5.125)	(0.173)	(64.09)	(0.392)	(0.0000496)
287(g) border	–61.10	–0.164***	3021	17.6**	–9,092	57.5***	0.00349
	(72.73)	(.0417)	(3,793)	(8.17)	(21,069)	(16.6)	(.00459)
Observations	21,403	4874	21,403	4923	21,403	4690	3681
Number of counties	-	1239	-	1239	-	1237	1237
Number of farms	11,501	-	11,501	-	11,501	-	-
				Panel B			
Aliens departed	-0.0917	–0.0092**	16.62**	0.572*	105.9	1.68**	–0.000209**
	(0.0948)	(0.00365)	(6.993)	(0.338)	(84.87)	(0.784)	(0.0000991)
287(g) border	–60.61	–0.155***	2931	17.2**	–9,662	56***	0.00375
	(73.04)	(0.0411)	(3,787)	(8.17)	(21,404)	(16.6)	(0.00463)
Observations	21,403	4874	21,403	4923	21,403	4690	3681
Number of counties	-	1239	-	1239	-	1237	1237
Number of farms	11,501	-	11,501	-	11,501	-	-
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C10
Impact of 287(g) authorization, task force model only: 2SLS with jail occupancy IV.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Labor expenses ($) (Census)	(4) Number of workers per county (Census)
287(g) authorization, task force model	–4.81***	306*	867**	–0.104**
	(1.71)	(168)	(367)	(0.0476)
287(g) border, task force model	–0.177***	21.3**	64.7***	0.0045
	(0.0527)	(9.78)	(19.7)	(0.00547)
County FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	4761	4809	4585	3579
Number of counties	1228	1229	1226	1202

a Heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C11
Impact of 287(g) authorization: counties with 287(g) programs in cities dropped, 2SLS.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
287(g) authorization	1996	-3.95***	107,536	254*	-3,580,000	763**	-.09**
	(4,556)	(1.35)	(266,690)	(139)	(4,885,000)	(313)	(.0393)
287(g) border	20.83	-.169***	7967	18.1**	-203,659	59.6***	0.0035
	(290.8)	(.0471)	(15,680)	(8.38)	(262,809)	(17.2)	(.00466)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	21,208	4860	21,208	4910	21,208	4676	3667
Number of counties	-	1239	-	1239	-	1237	1237
Number of farms	11,398	-	11,398	-	11,398	-	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C12
Impact of 287(g) authorization: control for urbanization.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Labor expenses ($) (Census)	(4) Number of workers per county (Census)
287(g) authorization	–2.92***	241*	701**	–.0743**
	(.869)	(131)	(281)	(.0348)
287(g) border	–.149***	19.9**	64.1***	0.00335
	(.0381)	(8.79)	(17.8)	(.00469)
Rural-urban continuum code:				
Metro, 250,000-1,000,000	–.0923**	–10.6	–72	–.00819
	(.0389)	(6.82)	(58.7)	(.00527)
Metro, below 250,000	–.108**	0.021	–122**	-.0154***
	(.0428)	(10.3)	(57.9)	(.00444)
Nonmetro, above 20,000, adjacent	–.16***	8.62	–123*	–.011
	(.0447)	(12.1)	(67.8)	(.00934)
Nonmetro, above 20,000, non-adjacent	–.209***	–29.3***	–239***	-.0266***
	(.0537)	(8.04)	(44)	(.00439)
Nonmetro, 2,500-20,000, adjacent	–.214***	–11.2	–207***	–.025***
	(.0435)	(7.23)	(43.4)	(.004)
Nonmetro, 2,500-20,000, adjacent	–.24***	–21.6***	–236***	–.0277***
	(.045)	(7.07)	(43.2)	(.00404)
Nonmetro, completely rural, adjacent	–.169***	–5.1	–178***	–.0224***
	(.0511)	(16)	(54.7)	(.00437)
Nonmetro, completely rural, non-adjacent	–.222***	–22.5***	–232***	-.0277***
	(.0513)	(8.08)	(44)	(.00407)
County FE	YES	YES	YES	YES
Farm FE	NO	NO	NO	NO
Year FE	YES	YES	YES	YES
Observations	4874	4923	4690	3681
Number of counties	1239	1239	1237	1237

a ”Adjacent” and ”non-adjacent” indicate whether a county shares a border with a metro county or not, respectively. b Heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses c Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. d *** p<0.01, ** p<0.05, * p<0.1 e Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C13
Impact of 287(g) authorization: control for crime rate and unemployment rate.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Labor expenses ($) (Census)	(4) Number of workers per county (Census)
287(g) authorization	–83**	13	8,165***	–0.229**
	(37)	(996)	(2,872)	(0.0891)
Violent crime rate	–0.0027	–0.0131	0.311	0.0000297**
	(.00244)	(.0553)	(0.453)	(.0000117)
Property crime rate	–0.00057	–.0438	–0.29	–0.000013
	(.0014)	(.0336)	(0.236)	(0.0000102)
Unemployment rate	0.0142	–6.23	–24.7	–0.00016
	(0.136)	(4.12)	(15.2)	(0.000761)
287(g) border	–3.6**	45	702**	–0.00528
	(1.46)	(71.7)	(318)	(0.00594)
County FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	4874	4923	4690	3681
Number of counties	1239	1239	1237	1237

a Heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures; Uniform Crime Reporting Data; BLS Local Area Unemployment Statistics

Table C14
Reduced form impact of jail occupancy on pre-treatment period outcomes.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Asset value per operation ($) (Census)	(4) Number of farms per county (Census)
Jail occupancy, pre-2002	0.0000007	0.00262***	0.0455*	-0.000
	(.0000158)	(.00084)	(.0275)	(0.000)
County FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Observations	2425	2443	2349	2450
Number of counties	1243	1243	1229	1244

a Jail occupancy constructed using data from years 1997–2002; outcomes are those available from the Census of Agriculture in the years 1997 and 2002. b Heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses c Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. d *** p<0.01, ** p<0.05, * p<0.1 e Source: US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C15
Impact of 287(g) authorization: All counties in control, 2SLS.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
287(g) authorization	–585.0	-2.64**	70,043*	343**	448,542	1,398***	-.13***
	(527.9)	(1.14)	(35,821)	(175)	(438,643)	(398)	(.0387)
287(g) border	–104.8***	-.089***	1305	14	–6,279	86.9***	0.00293
	(34.17)	(0.0333)	(4,152)	(9.35)	(15,626)	(21.5)	(0.00437)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	44,875	10,382	44,875	10,497	44,875	10,010	7845
Number of counties	-	2647	-	2647	-	2638	2645
Number of farms	24,299	-	24,299	-	24,299	-	-

a Standard errors robust to correlation at the strata level (ARMS) and heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C16
Impact of 287(g) authorization: 2SLS with random subsample.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Labor expenses ($) (Census)	(4) Number of workers per county (Census)
287(g) authorization	–3.701**	235.7	694.3**	-0.083*
	(1.435)	(150.8)	(328.8)	(0.042)
287(g) border	–0.190***	19.24*	62.83***	0.003
	(0.060)	(9.653)	(19.78)	(0.005)
County FE	YES	YES	YES	YES
Farm FE	NO	NO	NO	NO
Year FE	YES	YES	YES	YES
Avg. number of observations	3669	3706	3530	2771
Average number of counties	932.8	932.8	931.3	931.3

a Results reflect boot-strapped estimates of coefficients and standard errors from 100 iterations of the main regression in which 10% of the treated sample was randomly dropped in each iteration. b Heteroskedasticity-robust (Huber-White) standard errors (Census) in parentheses c Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. d *** p<0.01, ** p<0.05, * p<0.1 e Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C17
Impact of 287(g) authorization: 2SLS with ERS region clustered S.E.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
287(g) authorization	-446.7	-3.67*	80,960***	231	515,971***	686**	-.0809***
	(349.9)	(2.04)	(13,488)	(193)	(133,791)	(349)	(.0181)
287(g) border	-83.11	-.189***	7011	19.3	16,336**	62.9**	0.00274
	(92.66)	(.0587)	(4,987)	(16.1)	(7,549)	(29)	(.0048)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	21,403	4874	21,403	4923	21,403	4690	3681
Number of counties	-	1239	-	1239	-	1237	1237
Number of farms	11,501	-	11,501	-	11,501	-	-

a Wild cluster bootstrap t-stat for 287(g) authorization for Census specifications: (2) –2.46, (4) -0.37, (6) –0.47, (7) –1.84 b Standard errors robust to correlation at ERS resource region in parentheses c Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. d *** p<0.01, ** p<0.05, * p<0.1 e Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C18
Impact of 287(g) authorization: 2SLS with State clustered S.E.

	(1) Acres operated per farm (ARMS)	(2) Acres operated per county (Census)	(3) Fuel expenses ($) (ARMS)	(4) Fuel expenses ($) (Census)	(5) Labor expenses ($) (ARMS)	(6) Labor expenses ($) (Census)	(7) Number of workers per county (Census)
287(g) authorization	–446.7	–3.67***	80,960*	231	515,971	686**	-.0809***
	(458.3)	(1.33)	(47,667)	(199)	(661,954)	(324)	(.0311)
287(g) border	–83.11	–0.189***	7011	19.3	16,336	62.9***	0.00274
	(89.84)	(.048)	(4,416)	(11.8)	(25,436)	(23.7)	(.00466)
County FE	NO	YES	NO	YES	NO	YES	YES
Farm FE	YES	NO	YES	NO	YES	NO	NO
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	21,043	4874	21,403	4923	21,403	4690	3681
Number of counties	-	1239	-	1239	-	1237	1237
Number of farms	11,501	-	11,501	-	11,501	-	-

a Standard errors robust to correlation at the state level in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

results are largely consistent across these varied assumptions regarding standard error calculation. While more numerically intensive approaches are not possible for our farm-level data due to slow processing times through secure ARMS access, we are able to use estimate wild cluster bootstrap standard errors following Cameron and Miller (2015) with Census data, which are noted in Table C17.

Table C19
Impact of 287(g) authorization: 2SLS with County cluster SE.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Labor expenses ($) (Census)	(4) Number of workers per county (Census)
287(g) authorization	−3.67***	231*	686**	-.0809**
	(1.21)	(127)	(278)	(.0354)
287(g) border	−0.189***	19.3**	62.9***	0.00274
	(0.0514)	(8.4)	(17.2)	(0.00451)
County FE	YES	YES	YES	YES
Farm FE	NO	NO	NO	NO
Year FE	YES	YES	YES	YES
Observations	4874	4923	4690	3681
Number of counties	1239	1239	1237	1237

a Standard errors robust to correlation at the county level in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

Table C20
Impact of 287(g) authorization: 2SLS with Spatial SE.

	(1) Acres operated per county (Census)	(2) Fuel expenses ($) (Census)	(3) Labor expenses ($) (Census)	(4) Number of workers per county (Census)
287(g) authorization	−3.67***	231	686**	−0.0809**
	(1.3)	(150)	(309)	(0.0365)
287(g) border	−0.189***	19.3**	62.9***	0.00274
	(.0564)	(9.38)	(19.8)	(0.00451)
County FE	YES	YES	YES	YES
Farm FE	NO	NO	NO	NO
Year FE	YES	YES	YES	YES
Observations	4874	4923	4690	3681
Number of counties	1239	1239	1237	1237

a Spatial standard errors robust to correlation within 300 miles of the country centroid in parentheses b Census outcome variables are all scaled by the number of agricultural acres operated in the county in 2007 to account for differences in size and agricultural intensity across counties. c *** p<0.01, ** p<0.05, * p<0.1 d Source: USDA Agricultural and Resource Management Survey (ARMS); US Census of Agriculture (NASS QuickStats); BJS National Jail Census; ICE FOIA Proactive Disclosures

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